The Legibility of Income Risk

Sven van Holten Charria[†]

Abstract

What does the composition of a paycheck tell workers about their earnings risk? Recognising a temporary payment can help workers distinguish fluctuations that will reverse from lasting changes in pay. I examine whether this information helps explain why lower-income workers perceive greater and more persistent risk despite similar measured earnings variation across incomes. Using US household surveys, I show that bonuses account for a larger share of temporary variation and recur more regularly than other pay components at higher incomes, and that identifying them also yields greater gains in forecast accuracy. In a calibrated Bayesian learning model, information about temporary payment amounts and timing generates a modest decline in mean perceived risk with income and reproduces about half the persistence gap. Model experiments further show that identifying temporary payments or spreading transfers across months limits revisions to perceived risk, suggesting that payment design may influence consumption and saving through the information recipients draw from their payments.

*I am grateful to Ricardo Reis, Jonathon Hazell, Ben Moll, Marco Bellifemine, and participants of the LSE Pre-Doctoral seminars for their valuable comments and feedback, and I am especially indebted to my fellow pre-docs Nicholas Tokay, Elie Chadaida, and James Symons-Hicks for their willingness to keep on listening to my endless ideas.

[†]Pre-doctoral Research Assistant at the Centre for Macroeconomics, London School of Economics. E-mail: s.m.van-holten-charria@lse.ac.uk.

1 Introduction

Workers can experience similar fluctuations in pay yet draw very different conclusions about the security of their future earnings. Some can anticipate changes in pay or recognise temporary payments, while others must interpret movements in total pay without that information. What workers know about their payments can therefore shape the uncertainty they perceive and, in turn, how much they consume and save. This matters for models of household saving, which are commonly calibrated to earnings processes estimated from panel data. If those data do not capture the information workers use, a model can match observed earnings variation while misrepresenting households' uncertainty and their incentive to save.

In this paper, I study whether information about pay composition can help explain differences in perceived earnings risk across incomes. I compare earnings variation in the Survey of Income and Program Participation (SIPP) with workers' own assessments of future earnings risk in the New York Fed's Survey of Consumer Expectations (SCE). Lower-income workers in the SCE perceive greater risk over the coming year, conditional on keeping the same job, and their uncertainty is more persistent from one month to the next. The SIPP estimates do not show the same pattern: the variance of earnings changes over twelve months is approximately equal across income groups, and a proxy for earnings volatility is more persistent at higher incomes.

The SIPP's detailed payment records reveal differences in pay composition that may help explain these contrasting patterns. I primarily focus on bonuses, whose contribution to transitory earnings variation differs most across income groups and which account for approximately 36 percent at lower incomes and 68 percent at higher incomes. I establish three further empirical facts: (ii) these payments, such as performance or year-end bonuses, generally disappear in the following month, (iii) their annual recurrence is stronger at higher incomes, and (iv) identifying them provides information beyond that contained in total earnings histories. In simple forecasting regressions, including bonus shares alongside current and past total pay reduces three-month mean squared forecast error by 3.82 percent at higher incomes, compared with 0.40 percent at lower incomes. These gains come mainly from knowing the current bonus share and diminish at longer horizons. Higher-income workers therefore receive more variation through payment categories whose identification helps predict subsequent earnings, but whether this information helps explain the income patterns in perceived risk depends on how workers learn from their payments.

To answer this, I develop a Bayesian learning model in which workers infer both permanent pay and the variability of permanent and temporary earnings from successive payments. Holding their earnings histories fixed, I compare beliefs formed from total pay alone with those formed using additional information about temporary payment amounts and recurring payment dates. With known, constant variances, I show analytically how recognising temporary payments reduces uncertainty about permanent pay. I then simulate learning from successive payments using an earnings process calibrated to the SIPP, allowing workers to learn about earnings risk and examining both its perceived level and persistence. I choose the number of payments workers observe to target the SCE persistence estimates while leaving mean-risk ratios across income groups untargeted.

The calibrated model generates lower and less persistent perceived earnings risk at higher incomes but understates the size of these differences. Bonus recognition raises the lower–higher mean-risk ratio from 0.95 to 1.11, compared with 1.81 in the SCE. Higher-income workers experience more bonus variation, so identifying these payments reduces the variation they attribute to permanent shocks more strongly. Expected permanent shocks nevertheless dominate uncertainty over the twelve-month

forecast horizon, and their calibrated variances differ little across incomes, limiting the resulting differences in perceived risk. Allowing workers to also know payment dates helps explain the persistence gap: a worker forecasting earnings for her annual payment month includes uncertainty about the next bonus, but this contribution disappears when her forecast moves to the following month. Since annual bonuses are more common and more variable at higher incomes, these temporary increases in perceived risk make it less persistent there. I find that the benchmark generates a lower–higher persistence gap of 0.10, compared with 0.20 in the SCE, leaving the remaining income differences in both the level and persistence to be explained in future work.

Using the calibrated model, I examine what workers' inferences from temporary payments imply for the design of income support programmes. I start by showing that an unidentified one-month pay increase, such as a one-off transfer, leaves perceived risk higher even after earnings fall back because the worker interprets these movements as evidence of greater permanent earnings variability. If the transfer is recognised, the worker can distinguish it from changes in permanent pay, largely preventing this increase in perceived risk. Then, in a policy application, I compare paying a government transfer, such as a tax rebate, once a year with spreading the same total amount over twelve monthly instalments. Monthly delivery produces much smaller risk revisions when transfers are not distinguished from earnings. When each government receipt is identified, workers remove it before updating their earnings beliefs, and neither delivery schedule changes perceived earnings risk in the model. These results favour clearly identifying government transfers and suggest that spreading payments can limit risk revisions when recipients cannot distinguish them from earnings.

Taken together, the findings on how pay composition shapes perceived earnings risk have implications for consumption and saving. Differences in perceived risk can emerge from workers' payment histories even when they begin with the same beliefs, and a temporary receipt can change their desired protection against future earnings losses as well as their available resources. Accounting for this information may therefore change how consumption models distribute precautionary saving motives across households and predict spending responses to transfers. Quantifying those effects requires households' saving decisions and borrowing constraints to be modelled explicitly, and is left for future work.

1.1 Related Literature

Recent research shows that measured earnings variation and the risk households perceive can differ substantially. Caplin et al. (2026) and Tao Wang (2026) find that subjective risk is substantially below conventional estimates and examine the implications for saving. Caplin et al. (2026) also show that job transitions account for much of subjective earnings risk, whereas I focus on uncertainty about pay within the current job. Within this setting, I build on the evidence of Ben-David et al. (2018), who document greater uncertainty about income growth among lower-income households. I compare this income gradient with measured earnings variation and extend the comparison from levels to persistence, examining whether differences in the information workers receive about their pay components can explain both patterns.

Understanding what workers can learn from their pay requires distinguishing the sources of earnings variation. Lemieux, MacLeod and Parent (2009) show that performance pay contributes to rising wage inequality, particularly at the top of the distribution. Grigsby, Hurst and Yildirmaz (2021) further show that bonuses and overtime make total compensation more flexible than base wages. Whether these fluctuations create uncertainty also depends on their predictability. Closest to my empirical

analysis, Ganong et al. (2025) document frequent fluctuations in hours among hourly workers and temporary, often predictable bonus payments among salaried workers. I build on this evidence using the SIPP's separately reported payment categories to measure their contributions to transitory variation in earnings per hour within jobs. Beyond documenting their predictability, I test whether identifying a payment's components improves forecasts relative to observing only total earnings history.

The model builds on the income-learning framework of Druedahl and Jørgensen (2020), in which households distinguish persistent from transitory income shocks using earnings histories and private information. They infer almost perfect information about these components from annual income and consumption data. My model also allows workers to learn earnings variances, so recognising a current payment need not resolve their uncertainty about future risk. In a simpler model, Wang (2004) shows how uncertainty about income components can generate additional precautionary saving. This also connects to Orlik and Veldkamp (2024), who show that learning about an unknown distribution, particularly its tail probabilities, generates fluctuations in subjective uncertainty. I examine how information about temporary pay changes workers' learning about permanent and temporary earnings variances under a common updating rule.

Lastly, in the application to government transfers, I focus on households' information about these payments. Sahm, Shapiro and Slemrod (2012) find a higher reported propensity to spend stimulus delivered as a lump sum than through reduced tax withholding. Thakral and Tô (2025) show that households spend more after receipt when they have had less time to anticipate a transfer. Motivated by these findings, I examine how transfer recognition and delivery frequency affect perceived earnings risk, providing a potential channel through which payment design can influence precautionary saving and households' propensity to consume out of transfers.

2 Data and Measurement

The empirical analysis draws on two household surveys. I use the New York Fed's Survey of Consumer Expectations (SCE) to measure workers' uncertainty about their earnings over the coming year. The Survey of Income and Program Participation (SIPP) provides monthly earnings records, including separate amounts for bonuses, commissions, tips and overtime. I describe the two datasets and outline how perceived earnings risk and realised earnings variation differ across income groups, both in their levels and persistence.

2.1 Data on Earnings Expectations

The relevant question in the SCE asks employed respondents for the distribution of their earnings growth over the next twelve months, assuming that they remain in the same job, with the same employer and working the same hours.[1] In each monthly interview, respondents report this distribution by assigning probabilities to ten ranges of possible earnings growth, following the density-forecast approach of Engelberg et al. (2009). As in Wang (2026), I measure perceived earnings risk by the variance of this distribution, which I calculate from the resulting ten-bin histogram using bin midpoints.[2] The advantage of this question is that it asks workers to assess uncertainty about their pay while holding their job and hours fixed. Although I cannot verify how fully respondents follow these instructions, the elicited object is therefore closer to the conditional variance of the stochastic component of wage

[1] Appendix A reproduces the survey question and details the construction of perceived earnings risk.

[2] For the open-ended tails, I use ±16 percent, following the medium-width specification of Krüger and Pavlova (2024).

growth than to total income risk. In particular, it excludes the risk of losing or changing jobs, which Caplin et al. (2026) identify as an important source of subjective earnings risk. The analysis here therefore focuses on uncertainty about pay within the current job.

I restrict the sample to employees and require complete, valid histograms. I exclude responses that place all probability in a single bin and trim perceived variance at the pooled first and ninety-ninth percentiles. To follow workers' expectations within the same job, I drop each respondent's reports from the first interview in which she records a job change onwards and require at least six consecutive monthly reports. The resulting sample contains 35,697 reports from 3,790 respondents between August 2013 and February 2025, with a median of ten monthly reports per respondent. Since the two surveys interview different workers, I compare the risk reported by each income group in the SCE with earnings variation among workers in the corresponding SIPP group. I divide respondents into three groups by annual household income: below \$50,000, \$50,000–\$99,999, and at least \$100,000, which I refer to as lower, middle and higher income. The thresholds are fixed in nominal dollars over the sample. The SCE groups contain 6,746, 12,846 and 16,105 reports, respectively.

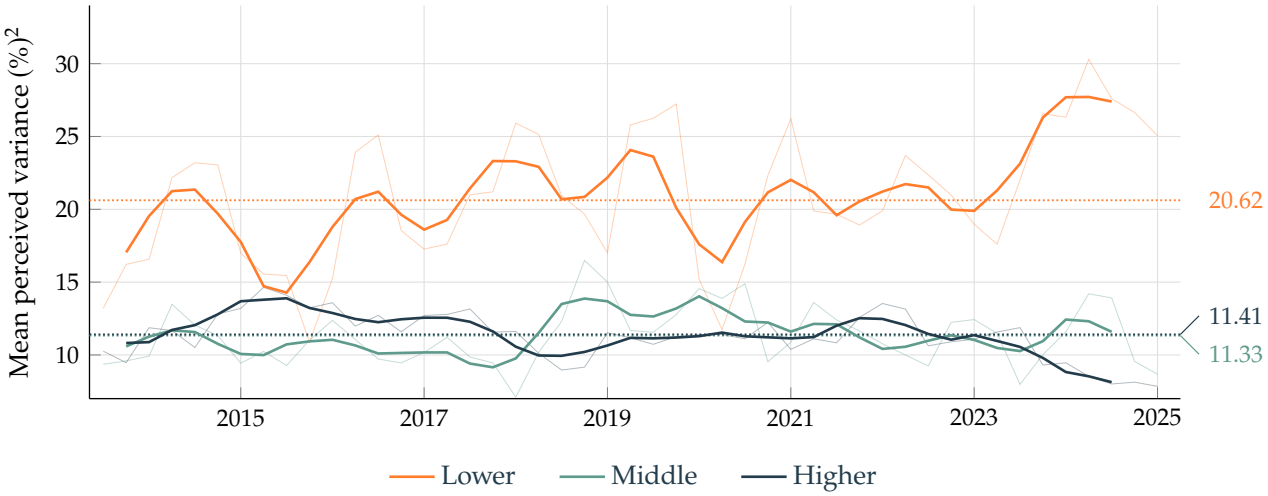


Figure 1: Perceived earnings variance by household-income group

Notes: SCE adjusted perceived variance, residualised, in squared percentage points. Thin lines show survey-weighted quarterly means; thick lines show their four-quarter centred moving averages. Dotted lines mark the survey-weighted means over the full sample. Lower-income workers report about 1.8 times as much perceived variance as higher-income workers.

To account for differences in demographics and interview dates across the income groups, I regress perceived variance on age, sex, education, state and calendar year-month effects. The regression also controls for the number of consecutive interviews completed, since repeated participation may affect reported beliefs, as Kim and Binder (2023) document for similar questions about inflation expectations in the SCE. Figure 1 shows that lower-income workers report greater earnings uncertainty, while the middle- and higher-income groups track each other closely. The lower–higher gap is present before and after adjusting for demographics, interview timing and survey participation, and also survives tighter income cut-offs (see Appendix A).

2.2 Data on Earnings and Pay Components

The SIPP is a longitudinal household survey conducted by the U.S. Census Bureau. Annual interviews collect employment and income histories for each month of the preceding calendar year. Alongside

total earnings, the survey records pay rates, hours worked, and separate amounts for bonuses, commissions, tips and overtime. These details allow me to examine both changes in a worker's total pay and the payments that account for them. I use earnings from the first recorded job, excluding self-employment and retaining uninterrupted periods of at least six months in the same job. I require positive earnings and reported hours. The underlying component panel contains 2,250,815 monthly observations for 103,134 workers over 2013–2023, with a median of seventeen observed months per worker.

Monthly earnings can change because a worker's pay changes or because she works more or fewer hours. Ganong et al. (2025) show that fluctuations in hours account for much of monthly earnings instability among hourly workers. To separate these sources of variation, I divide total monthly pay by reported monthly hours and study changes in hourly earnings. This measure includes the four additional payment categories, so a bonus can raise earnings per hour even when the worker's contractual pay rate is unchanged. For the comparison with perceived risk, I retain months in which reported hours change. Requiring unchanged hours would exclude a substantial share of earnings variation, particularly among lower-income workers, whose hours vary more (see Appendix Figure C.1). These payments may still inform workers' assessments of future pay, even when they are asked to hold future hours fixed. I therefore use the full same-job histories and report the unchanged-hours comparison in Appendix C.1.

Table 1: Annual earnings shares, receipt and size of pay components

	Lower	Middle	Higher
A. Mean share of annual earnings (%)			
Wages and salaries excluding itemised pay	96.65	96.52	95.80
Bonuses	0.26	0.44	1.24
Commissions	1.88	1.90	2.06
Tips	0.88	0.53	0.35
Overtime	0.33	0.61	0.56
B. At least one payment during the year (%)			
Bonuses	3.2	6.0	11.2
Commissions	2.6	3.2	3.9
Tips	2.5	1.5	1.0
Overtime	4.5	6.2	5.2
C. Mean monthly payment, conditional on receipt ($)			
Bonuses	985	1,978	10,150
Commissions	1,881	2,694	6,753
Tips	674	935	1,077
Overtime	396	708	1,228

Notes: Survey-weighted estimates, 2013–2023; amounts in nominal dollars. Panels A and B use complete same-job years; C conditions on receipt. Panel A averages workers' annual shares, including zeros. Hours may vary.

Before measuring earnings variation, I regress log hourly earnings on age, age squared, sex, race and education, with state and calendar year-month effects. I then study monthly changes in the residuals. For the variance decomposition, I exclude December-to-January changes because the two months are reported in separate annual interviews, creating a well-documented survey seam that produces artificial jumps in recorded earnings. I also trim squared residual changes at the pooled first and ninety-ninth percentiles. These adjustments do not ensure that the remaining changes are unexpected. For instance, a worker who knows that she will receive a scheduled pay rise faces no uncertainty about that increase, yet it can remain in residual earnings growth if the controls do not

capture its timing.

I separate monthly earnings into five components: the four itemised payments and the wages and salaries that remain after subtracting them from total earnings. Table 1 shows that wages and salaries excluding itemised pay account for about 96 percent of annual earnings in each income group. The mean annual bonus share rises sharply with income, from 0.26 to 1.24 percent, while the share from tips falls. This pattern is consistent with differences in the types of jobs held across income groups, with bonuses associated with professional and managerial roles and tips with service work. A payment can account for a small share of annual earnings and still cause a large monthly change if it is concentrated in a single month. Among workers observed in the same job for a complete calendar year, 3.2 percent of the lower-income group receive a bonus, compared with 11.2 percent of the higher-income group. Bonuses are also larger at higher incomes: in months when a bonus is received, the mean payment rises from \$985 to \$10,150, or from 22.5 to 30.4 percent of monthly pay. However, these means conceal substantial dispersion, with the median positive bonus in the higher-income group at just one-fifth of the mean. Appendix Table B.1 reports the long upper tail of bonus payments and the corresponding distributions of the other components.

2.3 Connecting the SCE and SIPP

To relate monthly earnings variation in the SIPP to workers' reported uncertainty in the SCE, I use a simple version of Meghir and Pistaferri's (2004) permanent–transitory framework, which separates lasting changes in pay from temporary fluctuations that may have disappeared by the SCE's twelve-month horizon.

The Earnings Benchmark. The standard process for adjusted log hourly earnings, z_{it}, is

$$z_{it} = p_{it} + \theta_{it}, \qquad p_{it} = p_{i,t-1} + \psi_{it}. \tag{1}$$

The permanent shock ψ_{it} accumulates through the random walk p_{it} and captures lasting changes in pay, such as the unanticipated part of a merit raise or cost-of-living adjustment. For a worker receiving such a raise, the increase carries through to her earnings in subsequent months. By contrast, a one-off discretionary bonus raises her earnings in the month of receipt without changing her regular pay rate. The transitory component θ_{it} captures these temporary fluctuations. Both shocks have mean zero, are i.i.d. across months and independent of each other, and have constant variances within each income group.

The permanent and transitory variances can be recovered from how earnings changes follow one another. A temporary payment raises earnings when it arrives and lowers them when it disappears, creating a negative relationship between consecutive changes. A permanent pay rise also raises earnings on receipt, but does not produce the same reversal the following month. The decomposition uses this difference to distinguish temporary fluctuations from lasting changes in pay. In particular, consider the monthly change $d_{it} = \psi_{it} + \theta_{it} - \theta_{i,t-1}$ and let $S_g = \mathbb{E}[d_{it}^2 \mid g]$ denote the second moment of monthly changes in income group g. Under the standard assumptions,

$$\mathcal{T}_g = -\mathbb{E}[d_{it} d_{i,t-1} \mid g], \qquad \mathcal{P}_g = S_g - 2\mathcal{T}_g, \tag{2}$$

where $\mathcal{T}_g$ is the temporary variance and $\mathcal{P}_g$ the permanent-innovation variance. I estimate these moments using survey-weighted averages of monthly changes and pairs of consecutive changes within

the same job, and allow reported hours to vary.

The SCE asks about earnings twelve months ahead, conditional on the same job, employer and hours. To compare this with the SIPP, I remove the predictable component of log hourly earnings and study changes in the residual, z_{it}.[3] This focuses the comparison on earnings movements around a predictable wage path. Under the benchmark process, the twelve-month change variance is

$$\text{Var}(z_{i,t+12} - z_{it} \mid g) = 12\mathcal{P}_g + 2\mathcal{T}_g. \tag{3}$$

Permanent shocks accumulate over the twelve months, while temporary pay enters through earnings in the initial and final months. The SIPP also shows dependence between earnings changes beyond one month, which the simple process does not capture (see Appendix Table C.2). The twelve-month variance is therefore a model-based benchmark.

Differences across Income Groups. In Table 2, I compare the SIPP and SCE measures of earnings risk. Permanent-innovation variation is 8.93 in the lower-income group and 8.39 in the higher-income group, while temporary variation rises from 2.05 to 4.06 ($p < 0.001$). This increase in temporary variation is consistent with Guvenen et al. (2021), who attribute rising annual earnings-growth dispersion among job-stayers at the top of the income distribution to temporary variation. Despite this increase, implied twelve-month earnings variation is similar across income groups ($p = 0.467$), whereas higher-income workers report less uncertainty in the SCE ($p < 0.001$).

Table 2: Perceived risk and realised earnings variation by income group

	SIPP				SCE
Income group	Monthly variation, S_g	Permanent $\mathcal{P}_g$	Transitory $\mathcal{T}_g$	Implied 12-month change variance	Perceived variance
Lower	13.02 (0.26)	8.93 (0.24)	2.05 (0.11)	111.21 (2.76)	20.62 (1.33)
Middle	14.75 (0.27)	8.64 (0.18)	3.06 (0.13)	109.75 (2.15)	11.33 (0.64)
Higher	16.51 (0.27)	8.39 (0.17)	4.06 (0.13)	108.76 (1.96)	11.41 (0.37)
Wald p-value	< 0.001	0.061	< 0.001	0.467	< 0.001

Notes: Survey-weighted estimates; hours may change within SIPP job spells. SCE variance is in squared percentage points; SIPP log variances are scaled by 10^4. Standard errors in parentheses cluster by respondent (SCE) or household (SIPP). Wald tests compare higher and lower incomes. See Appendix D.3.

At the twelve-month horizon, perceived variance amounts to 19 percent of the implied earnings-change variance in the lower-income group, compared with about 10 percent in both the middle- and higher-income groups. The proportional gap therefore widens with income. Wang (2026) similarly finds that wage-risk perceptions in the SCE fall well below conventional estimates from earnings panels due to measurement error in the earnings data. How much of this gap reflects mistaken beliefs remains unclear, since workers can anticipate payments that the earnings model treats as shocks (Pistaferri, 2001) and measurement error can distort recorded earnings. For instance, Bound and Krueger (1991) document serially correlated errors in earnings reports, so reporting error may affect both components of the decomposition. I therefore focus on relative differences in these ratios across income groups, assuming that reporting error inflates each group's SIPP benchmark by the same proportion.

[3] I obtain z_{it} as the residual from a Mincerian regression of log hourly earnings on age, age squared, sex, race and education, with state and calendar year-month effects. For the comparison with the SCE, I treat the fitted earnings change over the forecast horizon as known to the worker.

Persistence. Perceived risk is also more persistent within respondents at lower incomes. In particular, I examine whether a worker who reports more uncertainty than usual this month continues to do so in subsequent interviews. I estimate a first-order autoregression in demographic-adjusted perceived variance for each income group, with fixed effects for each uninterrupted response spell. Because the SCE follows respondents for only a short period, the usual fixed-effects estimator is subject to Nickell bias. I therefore use the system GMM estimator of Blundell and Bond (1998), which uses earlier reports as instruments for lagged perceived variance. Ma et al. (2020) similarly use dynamic-panel GMM to address short-panel bias when estimating persistence in managerial forecast errors. Table 3 reports a monthly persistence coefficient of 0.343 for the lower-income group, compared with 0.142 and 0.144 for the middle- and higher-income groups. The higher–lower difference is statistically significant ($p < 0.001$).

For the SIPP, I use a standard proxy for earnings volatility.[4] Its persistence measures whether large residual earnings movements, of either sign, tend to cluster over time. Using the same system GMM specification as for perceived risk, I find the opposite income gradient: the coefficient rises from 0.193 in the lower-income group to 0.235 in the higher-income group ($p = 0.007$).

Table 3: Monthly persistence of perceived risk and the SIPP volatility proxy

Income group	SIPP volatility proxy	SCE risk reports
Lower	0.193 (0.013)	0.343 (0.046)
Middle	0.187 (0.010)	0.142 (0.026)
Higher	0.235 (0.009)	0.144 (0.025)
Wald p-value	0.007	< 0.001

Notes: Monthly AR(1) coefficients from two-step system GMM. Standard errors in parentheses cluster by response spell and month (SCE) or job spell (SIPP). Wald tests compare higher and lower incomes. Appendices D.1–D.2 give the specification and supporting checks.

Taken together, these estimates show that higher-income workers report lower and less persistent perceived risk, despite greater earnings variation and a more persistent volatility proxy in the SIPP. The earnings records do not reveal which payments workers anticipated or recognised as temporary, however. For instance, a worker who recognises an unexpected bonus can distinguish it from a lasting change in regular pay, even though the payment still creates variation in her recorded earnings. The central question of this paper is therefore whether differences in workers' information about the components of their pay can help explain these opposing patterns. This requires understanding which payments generate the observed earnings variation and how workers might interpret and anticipate them.

3 Properties of Pay Components

The same increase in monthly earnings can have different implications for future pay depending on its source. For instance, a worker who recognises an increase as a bonus may expect her regular wage to remain unchanged, even if she expects the bonus to recur. Information about commissions

[4] I estimate $d_{it} = \alpha + \rho d_{i,t-1} + u_{it}$ by pooled, survey-weighted least squares, where $d_{it} = \Delta z_{it}$ is the adjusted monthly earnings change. The volatility proxy $\widehat{u}_{it}^2$ captures the size of movements not explained by the previous month's change, regardless of sign.

or occasional overtime may help her make a similar distinction. I compare the contributions of the different pay components and show that bonuses account for almost all of the increase in measured transitory earnings variation from lower to higher incomes. I therefore examine the duration and recurrence of these payments, with a focus on bonuses, to understand what workers could anticipate about these payments and what they could learn from them about future pay.

3.1 Contributions to Earnings Variation

I start by decomposing the transitory earnings variation reported in Table 2 into contributions from wages and salaries, bonuses, commissions, tips and overtime. I retain the same adjustment, trimming and same-job sample. Let d_{it}^k denote component k's contribution to the adjusted log earnings change. I construct these contributions so that they add up to the total change, $d_{it} = z_{it} - z_{i,t-1} = \sum_k d_{it}^k$.[5]

Each component's contribution includes its own earnings movements and its comovements with other payments. I split each cross-category term equally between the two categories. Using the same moment convention as in equation (2), component k receives

$$\begin{aligned} \mathcal{T}_{kg} &= -\frac{1}{2}\mathbb{E}\left[d_{it}^k d_{i,t-1} + d_{it} d_{i,t-1}^k \mid g\right] \\ &= \underbrace{-\mathbb{E}[d_{it}^k d_{i,t-1}^k \mid g]}_{\text{own-category term}} \underbrace{-\frac{1}{2}\sum_{\ell \neq k} \mathbb{E}\left[d_{it}^k d_{i,t-1}^\ell + d_{it}^\ell d_{i,t-1}^k \mid g\right]}_{\text{cross-category terms}}. \end{aligned} \tag{4}$$

The own-category term captures the tendency for a payment increase to be followed by a decline in the same component. The remaining terms account for movements across categories.

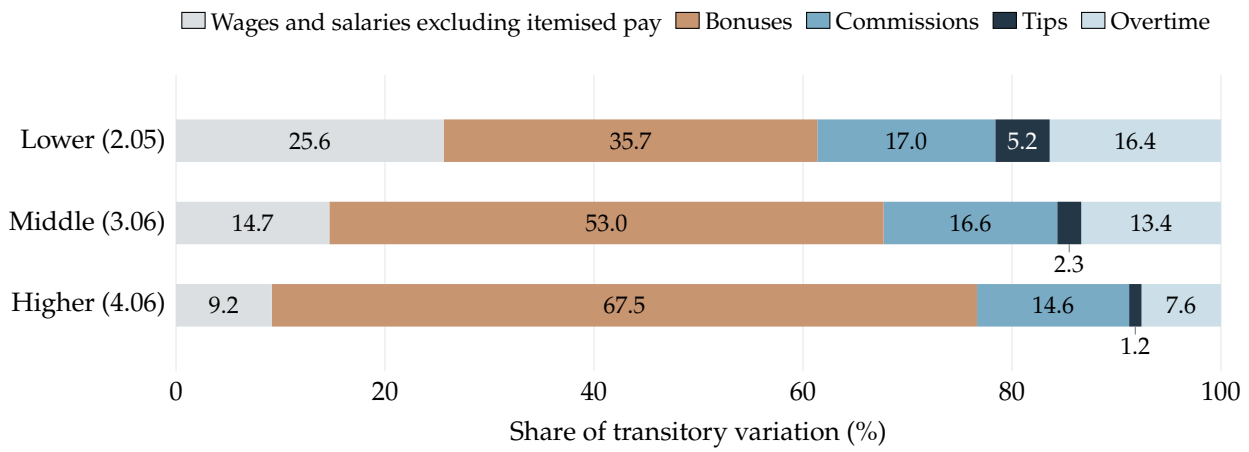


Figure 2: Pay components' shares of transitory variation

Notes: SIPP, 2013–2023, using the hours-may-change sample of Table 2. Shares include cross-category terms. Group totals appear in parentheses, with log moments scaled by 10^4.

Finding 1: Bonuses account for the increase in measured transitory earnings variation with income. Figure 2 shows that their share rises from 35.7 percent at lower incomes to 67.5 percent at higher incomes, a significant difference ($p < 0.001$). The bonus contribution rises from 0.73 to 2.74, while the combined contribution of the other four components remains close to 1.3 in both groups. If workers recognise

[5] I construct the contributions from changes in adjusted payment amounts, taking logarithms only of total pay. This allows months with zero bonuses or other payments to remain in the decomposition. Appendix E.1 details this construction.

bonuses as temporary, the greater bonus share at higher incomes may help them distinguish temporary payments from changes in permanent pay.

To check whether the shift towards bonuses extends to lasting earnings changes, I report the corresponding decomposition of permanent-innovation variation in Appendix Figure E.1. Wages and salaries excluding itemised pay account for at least three-quarters of permanent variation in every income group. The bonus share nevertheless rises from 4.7 percent at lower incomes to 12.4 percent at higher incomes. The bonus contribution therefore rises with income in both components, but the absolute increase is much larger for transitory variation.

3.2 Payment Duration

I next examine how the different pay components evolve after receipt, to understand what workers could learn from the source of an earnings increase. For instance, a worker who recognises a spike in pay as a bonus may expect it to disappear without revising her view of regular pay. When additional payments continue for several months, distinguishing them from a lasting pay rise using total earnings alone may be harder.

To identify the start of a payment spell, I focus on receipt months preceded by at least two consecutive months with no contribution to total earnings from the relevant component. This helps distinguish a new overtime spell, for instance, from one already under way. The two-month rule excludes restarts after a single missed month while still admitting quarterly payments. As a robustness check, I repeat the analysis using either one or three prior months without receipt and find that bonuses still fall away quickly while overtime remains positive through month six (see Appendix Table F.1).

I estimate local projections to examine how each component's current payment relates to its payments over the next six months. For component k, let $x^k_{i,t+h}$ denote its hourly amount in month $t+h$, scaled by the worker's average total hourly earnings over the twelve months before t. Holding this scale fixed across horizons, I estimate

$$x^k_{i,t+h} = \beta^k_h x^k_{it} + C'_{it}\gamma^k_h + \tau^k_{t,h} + \alpha^k_{g,h} + \varepsilon^k_{i,t+h}, \qquad h = 0, \ldots, 6. \tag{5}$$

where C_{it} contains twelve months of earnings and payment history, log average prior hourly earnings and log hours in the preceding month. The calendar-date effects $\tau^k_{t,h}$ account for shared payment timing, such as year-end bonuses, while the income-group effects $\alpha^k_{g,h}$ allow average payments to differ across groups. After accounting for these patterns and the worker's prior payment history, β^k_h measures whether a larger payment today is followed by larger payments h months later.

Finding 2: Bonuses fall away quickly, commissions show quarterly recurrence, and overtime payments persist for several months. In Figure 3, I show that the bonus coefficient is close to zero following an initial payment, with small positive coefficients at months three and six, suggesting that bonuses mainly provide one-off additions to monthly pay. Commissions instead display pronounced peaks of 0.45 and 0.39 at these horizons, consistent with quarterly payments: the additional pay can disappear between payment dates and return at regular intervals. For overtime, the coefficient remains 0.58 after one month and 0.25 after six months. This gradual decline is consistent with spells of extra work that sustain additional earnings for several months. Too few workers begin receiving tips after a payment gap to estimate their duration reliably. I extend the profiles to twelve months in Appendix Figure F.1, which shows a pronounced annual bonus peak (0.433, SE 0.041). The estimates for commissions and

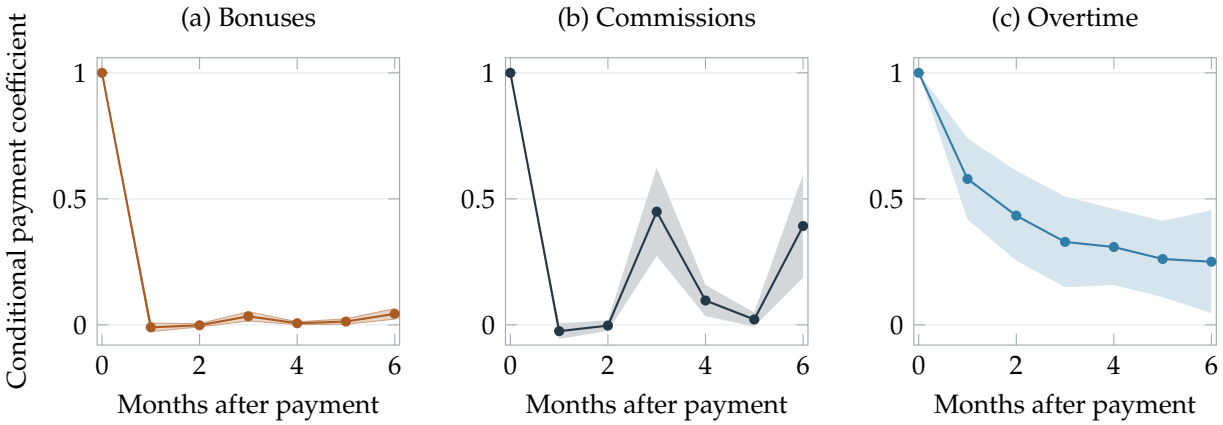


Figure 3: Payment profiles following two months without receipt

Notes: SIPP, 2013–2023. I pool income groups because there are too few new payment episodes to estimate each component's path precisely within every group. Workers remain in the same recorded job throughout the observation window, while their hours may change. Coefficients relate future to current normalised hourly payments; the initial coefficient is one by construction. Shading gives 95% pointwise confidence intervals with standard errors clustered by worker.

overtime at this horizon are less precise, with overtime only just significant at the 5% level.

These payment profiles suggest that knowing the source of an earnings increase may help workers assess whether their regular pay has changed. For instance, a worker who recognises a bonus as temporary can set it aside when assessing her regular wage, even if she did not anticipate receiving it. Overtime can instead sustain earnings for several months without a permanent pay rise, making the distinction harder to infer from total pay alone. Since bonuses account for more temporary earnings variation at higher incomes, recognising their temporary nature may be particularly valuable for these workers.

3.3 Annual Bonus Recurrence

Bonus history may also help workers anticipate future payments. I examine whether last year's bonus helps predict both the receipt and amount of a payment in the same month this year. For instance, a worker who regularly receives a December bonus may anticipate the rise in pay that month and the fall afterwards, while remaining uncertain about whether the payment will arrive and how large it will be. These anticipated earnings movements still contribute to measured variation, even though they need not generate uncertainty for the worker.

For a bonus receipt, I estimate the following regression separately within each income group g, with B_{it} indicating whether a bonus was received:

$$B_{it} = c_{i,j(i,t)} + \tau_t + \gamma_g B_{i,t-12} + \eta_g B_{i,t-1} + \zeta_g F_{it} + \varepsilon_{it}, \tag{6}$$

where $F_{it} = \frac{1}{11}\sum_{s=1}^{11} B_{i,t-s}$ controls for bonus receipt frequency over the preceding eleven months, and $B_{i,t-1}$ controls for receipt last month. Together, these help distinguish annual recurrence from bonuses paid frequently or in consecutive months. The worker–job effects $c_{i,j(i,t)}$ account for some jobs routinely paying bonuses and others rarely doing so, while the calendar-date effects τ_t account for common payment timing, such as bonuses being more frequent in December. The annual coefficient γ_g then measures whether a worker who for instance received a bonus last December is more likely to receive one this December, beyond the general increase in bonus receipt that month and conditional

on her job and recent payment history.

I repeat the regression using monthly bonus amounts, replacing the receipt indicators with amounts and F_{it} with their preceding eleven-month average.

Table 4: Annual recurrence in bonus receipt and amounts

Association with last year's bonus	Lower	Middle	Higher	Higher–lower p-value
Receipt probability (pp) per receipt last year	6.9 (3.4)	7.5 (1.8)	20.6 (1.3)	< 0.001
Amount (\$) per \$100 more last year	2.75 (2.29)	7.34 (5.61)	29.83 (3.22)	< 0.001

Notes: Survey-weighted regressions with worker-clustered standard errors in parentheses. Wald tests compare higher- and lower-income annual coefficients. Amounts are in nominal dollars. Both exercises require a complete preceding year in the same recorded job and allow hours to change.

Finding 3: Bonus receipt shows stronger annual recurrence at higher incomes. Table 4 shows that receipt in the same month last year is associated with a 6.9 percentage-point increase in current receipt probability at lower incomes, compared with 20.6 points at higher incomes ($p < 0.001$ for the difference). Commissions, tips and overtime do not show the same positive annual association in receipt after controlling for recent payment history (see Appendix Table G.1, panel A). For bonuses, the higher-income advantage is specific to the twelve-month lag among the horizons examined. At three months, recurrence is slightly stronger at lower incomes, while at six months the higher–lower difference is not statistically significant (see Appendix Table G.2). The pattern therefore points to annual timing as a distinctive feature of bonus receipt at higher incomes.

The baseline amount regressions show a similar annual gradient: an additional \$100 in last year's same-month bonus is associated with \$2.75 more today at lower incomes and \$29.83 more at higher incomes. The amount regressions for commissions, tips and overtime likewise show no positive annual association after controlling for recent payment history (see Appendix Table G.1, panel B). One possible explanation, consistent with the payment profiles in Figure 3, is that overtime follows spells of extra work lasting several months without recurring annually, while commissions follow shorter, quarterly payment cycles, but ultimately don't contribute to the annual result.

Table 5: Bonus timing and earnings variation

	Lower	Middle	Higher	Higher–lower p-value
Months with bonus receipt one year earlier (%)	0.58 (0.07)	1.14 (0.07)	1.62 (0.07)	< 0.001
Share of bonus variation from these months (%)	17.9 (6.4)	33.7 (4.3)	32.5 (2.3)	0.032
Share of total temporary variation from bonuses in these months (%)	6.6 (2.4)	19.4 (3.0)	23.5 (1.9)	< 0.001

Notes: Survey-weighted estimates allowing hours to change, with the required same-job payment history. Worker-clustered analytical standard errors are in parentheses. Wald tests compare higher and lower incomes. Appendix G.4 describes the calculation.

In Table 5, I show that past bonus receipt identifies months in which bonuses account for a substantial share of temporary earnings variation. At higher incomes, months with bonus receipt in the same job one year earlier represent only 1.6% of eligible observations, yet their bonus contribution accounts for 23.5% of total temporary variation. At lower incomes, the corresponding shares are 0.6% and 6.6%. These results motivate a separate bonus component that can recur in the same calendar month each year. For instance, a worker who receives a bonus this month may expect another twelve months ahead, at the SCE's forecast horizon. Her uncertainty about future earnings then depends on what her bonus history tells her about the receipt and size of that payment, as well as uncertainty about the rest of her pay.

3.4 Forecasting Earnings with Bonus Information

Lastly, I test whether knowing how much of a worker's pay comes from bonuses improves forecasts of her future earnings beyond the information in total pay. For instance, a worker who receives an unusually high payment may infer a smaller rise in regular pay if she knows that part of it is a bonus. To provide an intuitive link, I use a simple linear regression to compare forecasts based on current and past total earnings with forecasts that also include the bonus share of those payments. For log hourly earnings w_{it}, I estimate the following regression separately by income group g and forecast horizon h:

$$w_{i,t+h} - w_{it} = \alpha_{gh} + X_{it}'\gamma_{gh} + \boldsymbol{c}_{it}'\pi_{gh} + u_{it,h}, \qquad h \in \{3, 6, 12\}. \tag{7}$$

The baseline regression omits $\boldsymbol{c}_{it}$ and uses only X_{it}: current log hourly earnings, their three monthly lags, their average over earlier observations in the same job, and calendar-month and year controls. The additional variables $\boldsymbol{c}_{it}$ contain the current bonus share, its three monthly lags and its average over earlier observations in the same job. Each share divides bonus pay by total pay in that month. The comparison therefore asks whether the payment breakdown adds information beyond the level and recent path of total earnings.

Since bonus variables can improve fit simply by capturing noise, I split workers into five sets and predict each using regressions estimated on the other four. Both specifications use the same observations and survey weights, and I measure improvement as the percentage reduction in mean squared forecast error.

Table 6: Forecast gains from bonus information

Forecast horizon	Lower	Middle	Higher	Higher–lower p-value
3 months	0.40 (0.12)	0.96 (0.12)	3.82 (0.18)	< 0.001
6 months	0.14 (0.06)	0.38 (0.06)	1.81 (0.11)	< 0.001
12 months	−0.01 (0.01)	−0.04 (0.04)	0.21 (0.06)	< 0.001

Notes: Percentage reductions in survey-weighted mean squared forecast error. Approximate worker-clustered standard errors in parentheses condition on fitted forecasts; Wald tests compare higher and lower incomes. Workers remain in the same recorded job; hours may change. Samples vary by horizon. Appendix H.1 gives further details.

Finding 4: Bonus information improves earnings forecasts more at higher incomes, particularly over the next three to six months. Table 6 shows that adding bonus information reduces three-month mean squared forecast error by 3.82 percent at higher incomes, compared with 0.40 percent at lower incomes ($p < 0.001$). The

gain declines with the forecast horizon but remains significant at higher incomes. At twelve months, it amounts to only 0.21 percent, with no significant improvement in the other two groups. When I add the other pay components at this horizon, the forecast gain at higher incomes rises from 0.21 to 0.43 percent ($p = 0.025$ for the additional gain), suggesting that information about future earnings extends beyond bonuses, although the annual gains remain modest (see Appendix Table H.2).

To separate the contributions of current and past bonus information, I also estimate forecasts that add only past bonus shares to the total-pay baseline. These produce little improvement over three and six months, with most of the gain appearing when I also include the current bonus share (see Appendix Table H.1). These near-term gains are consistent with current bonus information helping workers distinguish temporary payments from changes in regular pay. Whether this information helps explain differences in perceived risk across incomes depends on how workers use it, which the forecast comparison alone does not establish and I therefore examine in a learning model.

4 A Model of Learning about Risk

I have shown that bonus information improves earnings forecasts, particularly at higher incomes. To examine how information about temporary payments affects workers' earnings uncertainty, I now develop a simple Bayesian learning model. The worker observes total earnings and may separately identify part of her temporary pay and know when some of these payments recur. Bonuses motivate this information structure, but it can also describe other temporary payments under the same assumptions. I use the Kalman filter as a tractable framework for describing how workers learn about permanent pay and how uncertainty about its level contributes to perceived earnings risk. I first show analytically how identifying a larger share of temporary pay affects workers' forecast uncertainty in a simplified setting where variances are known and constant. I then allow workers to learn these variances and use the quantitative model to examine differences across income groups in the level and persistence of perceived earnings risk.

4.1 Environment

Earnings Process. I retain the permanent–transitory structure of the earnings benchmark and, crucially, distinguish temporary pay that the worker can separately identify, such as bonuses, from the remaining temporary pay:

$$w_t = p_t + u_t + b_t, \qquad p_t = p_{t-1} + \psi_t, \tag{8}$$

where w_t denotes adjusted log hourly earnings in month t, p_t is permanent pay, b_t is a temporary component that the worker can identify separately and u_t captures the remaining temporary pay. Bonuses provide an example of a payment component that the worker can observe separately and recognise as temporary, though the framework also applies to other payments with these characteristics, like temporary transfers. The shock ψ_t permanently changes pay, while u_t and b_t affect earnings only in the month received. In the baseline, the temporary components are centred Gaussian terms, $u_t \sim N(0, \sigma_u^2)$ and $b_t \sim N(0, \sigma_b^2)$, with constant variances. Their one-month duration approximates the short-lived bonus payments in Figure 3.

For parsimony, I abstract from recurrence in other payments and the longer duration of overtime, since their combined contribution to transitory variation changes little across income groups. I leave modelling these dynamics for future work. For analytical tractability, the baseline also excludes annual bonus recurrence, which I introduce below and examine in the quantitative model in Section 5.

I allow permanent-pay risk to vary over time through a monthly version of the ARCH specification in Meghir and Pistaferri (2004), where q_t denotes the variance of the permanent shock:

$$\psi_t = \sqrt{q_t}\,\varepsilon_t, \qquad q_{t+1} = (1-\alpha)\bar{q} + \alpha\psi_t^2, \qquad \varepsilon_t \sim N(0,1), \tag{9}$$

where $\bar{q} > 0$ is the long-run mean of q_t and $0 \leq \alpha < 1$. For $\alpha > 0$, a worker who experiences a large permanent pay rise or cut faces greater risk of another large change, although its direction remains uncertain. A higher α makes this elevated risk more persistent; at $\alpha = 0$, risk is constant at $\bar{q}$. The shocks ε_t, u_t and b_t are independent across components and months, and independent of the initial states, conditional on the parameters.

For the quantitative analysis, I extend the baseline to allow bonus payments to recur annually, motivated by the annual bonus peak in Appendix Figure F.1 and the recurrence estimates in Section 3.3. The worker may then know when a temporary payment will arrive, while remaining uncertain about its amount. The same idea can describe what workers know about other scheduled payments, such as government transfers or holiday pay. To model this recurrence, I split bonus pay into irregular monthly bonus variation and an annual payment, writing $b_t = v_t + D_t a_t$, with $v_t \sim N(0, \sigma_v^2)$ and $a_t \sim N(0, \sigma_a^2)$. I let an empirically calibrated share ρ of workers receive an annual payment. For these workers, $D_t = 1$ in their annual payment month and zero otherwise. For the remaining workers, it is always zero. I retain the Gaussian and independence assumptions of the baseline for ε_t, u_t, v_t and a_t, conditional on the parameters and payment schedule.

Worker Information and Perceived Risk. I build on the income-learning framework of Druedahl and Jørgensen (2020) and assume that the worker knows the form of the baseline earnings process but observes neither permanent pay p_t nor its risk q_t. She infers both from her payments and learns the fixed parameters $\vartheta = (\bar{q}, \alpha, \sigma_u^2, \sigma_b^2)$. Workers start with a common prior over ϑ, so income differences in beliefs are not imposed at the outset.[6] The true parameters differ across groups, and workers learn about them from their own payments using Bayes' rule.

I first compare two nested information sets with common initial information $\mathcal{I}_0$. Observing total pay gives the worker $\mathcal{I}_t^0 = \mathcal{I}_0 \cup \{w_s\}_{s=1}^t$, under which she uses the signal $y_t^0 = w_t$. Observing the temporary component separately expands her information to $\mathcal{I}_t^1 = \mathcal{I}_t^0 \cup \{b_s\}_{s=1}^t$, allowing her to subtract this payment and use $y_t^1 = w_t - b_t = p_t + u_t$. She still has to distinguish permanent pay from the remaining temporary component u_t, but faces less noise in doing so.

In the annual-payment exercise, the worker may also know her payment schedule. I denote this information by $\mathcal{I}_t^2 = \mathcal{I}_t^1 \cup \{C\}$, where $C = 0$ denotes no annual payment and $C \in \{1, \ldots, 12\}$ its payment month. She then learns the variances of irregular and annual bonuses, σ_v^2 and σ_a^2. Without calendar knowledge, she estimates a single monthly bonus variance and does not infer a schedule from past receipts. The numerical calendar comparison therefore changes her forecasting model as well as her information.

I measure perceived risk by the worker's conditional variance of earnings twelve months ahead, matching the SCE forecast horizon. Iterating the permanent-pay process in equation (8) forward gives

$$w_{t+12} = p_t + \sum_{h=1}^{12} \psi_{t+h} + u_{t+12} + b_{t+12}.$$

[6] Conditional on the parameters, initial permanent pay has a Gaussian prior and initial risk is drawn from the long-run distribution implied by the ARCH process. The prior gives initial pay a finite second moment and the variance parameters finite means, ensuring that earnings forecast variances are well defined.

The worker is therefore uncertain about current permanent pay, the permanent shocks that accumulate over the coming year, and temporary payments received at the forecast date. Under the baseline assumptions, these components have zero conditional covariance, so their variances add. For $j \in \{0, 1\}$, perceived earnings risk is therefore:

$$V^j_{t,12} = \text{Var}(w_{t+12} \mid \mathcal{I}^j_t) = P^j_t + \mathbb{E}[\sigma^2_u + \sigma^2_b \mid \mathcal{I}^j_t] + \sum_{h=1}^{12} \mathbb{E}[q_{t+h} \mid \mathcal{I}^j_t], \tag{10}$$

where $P^j_t = \text{Var}(p_t \mid \mathcal{I}^j_t)$ measures uncertainty about current permanent pay. The future shock variances enter as conditional expectations because the worker is uncertain about the variance parameters and future risk. With a known payment schedule, the bonus term instead includes irregular variation and, when an annual payment is due, its variance. Appendix I.1 formally derives the baseline decomposition and this extension.

4.2 Bayesian Learning

The worker observes a sequence of payments and must infer current permanent pay and learn how much permanent and temporary pay fluctuate. These inferences are linked: her beliefs about earnings risk affect how she interprets each payment, while the permanent change she infers provides evidence about future risk. I first explain inference about permanent pay with known, constant variances, then allow the worker to learn about earnings risk.

Learning about Permanent Pay. To explain the Kalman update in the baseline without recurring payments, I temporarily assume that the worker perfectly knows the variance parameters and set $\alpha = 0$, so $q_t = \bar{q}$. This means that she knows how much permanent and temporary pay fluctuate, but still has to infer how much of each payment is permanent. I relax the known-variance assumption when turning to learning about earnings risk.

I follow the standard Kalman filtering procedure. Before the payment at date t, she expects permanent pay to equal her previous estimate, $\widehat{p}^j_{t-1} = \mathbb{E}[p_{t-1} \mid \mathcal{I}^j_{t-1}]$, since the new permanent shock has mean zero. The shock nevertheless adds uncertainty, so the prediction variance is $P^j_{t|t-1} = P^j_{t-1} + \bar{q}$, where $P^j_{t-1} = \text{Var}(p_{t-1} \mid \mathcal{I}^j_{t-1})$ measures her remaining uncertainty about last month's permanent pay. The payment then provides a noisy signal of permanent pay. With total-pay information, the temporary noise has variance $R_0 = \sigma^2_u + \sigma^2_b$; recognising the temporary component reduces it to $R_1 = \sigma^2_u$. On observing y^j_t, she combines this signal with her previous estimate using the Kalman update:

$$\widehat{p}^j_t = \widehat{p}^j_{t-1} + K^j_t \underbrace{(y^j_t - \widehat{p}^j_{t-1})}_{\text{pay surprise}}, \qquad K^j_t = \frac{P^j_{t|t-1}}{P^j_{t|t-1} + R_j}, \qquad P^j_t = \underbrace{(1 - K^j_t)}_{\text{share retained}} P^j_{t|t-1}. \tag{11}$$

Here $\widehat{p}^j_t$ is her revised estimate of permanent pay, P^j_t measures her remaining uncertainty about it, and $e^j_t = y^j_t - \widehat{p}^j_{t-1}$ denotes the pay surprise. The Kalman gain K^j_t is the weight she places on the pay surprise: she relies more on the payment when her previous estimate is uncertain, and less when temporary noise makes the payment harder to interpret. For a given prediction variance, recognising temporary payments lowers R_j, raises K^j_t and hence reduces P^j_t.

Information about temporary payments also changes the pay surprise itself. For instance, a worker who can attribute an unusually high payment to a bonus has little reason to revise her estimate of

permanent pay upwards. Overall, recognising the bonus can make the worker more certain about permanent pay without requiring a large revision to her estimate of its level.

Learning about Earnings Risk. I now allow the worker to learn the variance parameters and return to the time-varying risk process in equation (9). Since a large permanent pay rise or cut raises the risk of another large change when $\alpha > 0$, the worker's interpretation of a pay surprise also informs her assessment of future shock variances, affecting both the level and persistence of perceived earnings risk.

After observing each payment, she uses Bayes' rule to update her beliefs about the permanent shock and the parameters governing the earnings process. Since she knows the ARCH structure, she obtains her estimate of next month's permanent-shock variance, $\widehat{q}^{\,j}_{t+1|t} = \mathbb{E}[q_{t+1} \mid \mathcal{I}^j_t]$, by averaging the ARCH equation over her updated beliefs:

$$\widehat{q}^{\,j}_{t+1|t} = \mathbb{E}\left[(1-\alpha)\bar{q} + \alpha\psi_t^2 \mid \mathcal{I}^j_t\right]. \tag{12}$$

Because the permanent shock is unobserved, she must account for both her estimate of its size and her uncertainty about it. The identity $\mathbb{E}[\psi_t^2 \mid \mathcal{I}^j_t] = (\mathbb{E}[\psi_t \mid \mathcal{I}^j_t])^2 + \mathrm{Var}(\psi_t \mid \mathcal{I}^j_t)$ shows why using only the squared estimate would omit relevant uncertainty. For instance, a worker whose estimated permanent shock is close to zero may still consider sizeable positive and negative shocks plausible. Her expected squared shock can therefore remain large even when her estimated change in permanent pay is small.

Information about temporary payments therefore changes what the worker learns about earnings risk. Observed payment amounts inform her beliefs about σ_b^2, while subtracting them helps her assess permanent shocks from the remaining pay. For instance, recognising an unusually large payment as a bonus can weaken the evidence that permanent pay has become riskier. Over successive payments, this distinction also affects what she learns about the long-run permanent variance $\bar{q}$ and the persistence of risk α. Appendix J.1 gives the joint update used in the numerical calculation.

Learning about Scheduled Payments. Under $\mathcal{I}_t^2$, the worker knows when her annual payment arrives. Outside that month, she observes only irregular bonus variation and learns about σ_v^2. In the annual payment month, her bonus includes both components. By accounting for the irregular variation, she also learns about annual-payment variance, σ_a^2. A large annual payment can therefore raise her assessment of annual-payment variability without implying the same increase in bonus risk throughout the year.

Knowing the date then helps her assess which future payments matter for her forecast. For instance, a worker expecting a December bonus includes uncertainty about next December's payment in her December report. In January, she forecasts the following January, which contains no annual payment. Her bonus contribution to twelve-month perceived risk is therefore

$$B^2_{t,12} = \mathbb{E}[\sigma_v^2 \mid \mathcal{I}_t^2] + D_{t+12}\mathbb{E}[\sigma_a^2 \mid \mathcal{I}_t^2]. \tag{13}$$

The bonus contribution to reported risk therefore falls when the forecast month no longer includes an annual payment, even if the worker's beliefs about bonus variability remain unchanged. I examine how much this contributes to the persistence gradient in the quantitative model, where the worker learns jointly about permanent pay, earnings risk and the parameters.

4.3 Implications for Perceived Earnings Risk

I next examine analytically how pay information affects the level of perceived earnings risk and the persistence of errors in workers' risk assessments.

The Level of Perceived Earnings Risk. For the baseline earnings process in Section 4.1, I derive analytical results showing how information about temporary pay can improve earnings forecasts, lower perceived risk and affect risk updating. I first express the contribution of future permanent shocks in terms of current risk, its long-run level and its persistence.

Proposition 1 (Permanent-risk persistence and forecast uncertainty)**.** *Under the baseline earnings process in Section 4.1, twelve-month perceived earnings risk is*

$$\begin{aligned} V^j_{t,12} = P^j_t + \mathbb{E}[\sigma^2_u + \sigma^2_b \mid \mathcal{I}^j_t] + 12\mathbb{E}[\bar{q} \mid \mathcal{I}^j_t] \\ + \mathbb{E}\Big[A_{12}(\alpha)(q_{t+1} - \bar{q}) \mid \mathcal{I}^j_t\Big], \end{aligned} \tag{14}$$

where $A_{12}(\alpha) = \sum_{h=0}^{11} \alpha^h$ is the cumulative weight on current risk over the twelve-month horizon. Appendix I.1 derives this decomposition.

Permanent shocks accumulate over the year, giving the third term, while the final term adjusts for whether current permanent risk is above or below its long-run level. A higher α makes this deviation last longer and therefore gives it more weight in the twelve-month forecast. Since recognising today's payment helps distinguish temporary pay from permanent shocks, it can affect both uncertainty about current permanent pay and expected future permanent risk. Recognising today's payment does not reveal the amounts of future temporary payments, which still contribute to her earnings uncertainty.

Proposition 2 (Pay information and perceived earnings risk)**.** *Conditional on the total-pay history, observing the history of b_t weakly reduces expected perceived earnings risk under a common earnings process and prior:*

$$V^0_{t,12} - \mathbb{E}[V^1_{t,12} \mid \mathcal{I}^0_t] = \mathrm{Var}(\widehat{p}^1_t \mid \mathcal{I}^0_t) \geq 0. \tag{15}$$

The reduction is strict whenever the additional information changes the forecast of permanent pay with positive conditional probability. The result follows from the law of total variance, as shown in Appendix I.2.

The right-hand side measures how much the forecast of permanent pay differs across histories of b_t consistent with the same total-pay history. For instance, learning which payments were bonuses removes this source of uncertainty on average; the result need not hold after every particular bonus observation.[7] The worker's forecast variance is also her expected squared forecast error when she uses her posterior mean to predict earnings. This provides a theoretical counterpart to Finding 4, where bonus information improves earnings forecasts outside the estimation sample. The information comparison also extends to recurring payments and to knowledge of the payment schedule (see Appendix I.2).

I next examine how the share of temporary variation the worker can identify affects her earnings uncertainty. To isolate the role of pay composition, I hold permanent and total temporary variance fixed and vary the share the worker can identify.

[7] The average reduction in forecast uncertainty follows more generally from nested information sets and does not depend on the information concerning bonuses.

Proposition 3 (Pay composition and the level of perceived risk)**.** *Suppose the variance parameters are known, there are no recurring payments and $\alpha = 0$. In the steady state of the Kalman filter,*

$$\begin{aligned} P^j_\infty &= \frac{\sqrt{\bar{q}^2 + 4\bar{q}R_j} - \bar{q}}{2}, \\ V^j_{\infty,12} &= P^j_\infty + 12\bar{q} + \sigma_u^2 + \sigma_b^2, \end{aligned} \tag{16}$$

where $R_0 = \sigma_u^2 + \sigma_b^2$ and $R_1 = \sigma_u^2$. Holding $\bar{q}$ and positive total temporary variance R_0 fixed, a larger separately observed share of temporary variance strictly lowers $V^1_{\infty,12}$ and leaves $V^0_{\infty,12}$ unchanged. Appendix I.3 derives the steady-state variance and establishes the effect of pay composition.

For instance, two workers can experience equally variable earnings, yet the worker who can identify more of the temporary variation as bonuses faces less noise when estimating permanent pay. Here, future shocks contribute the same risk, so the difference must then come from uncertainty about current permanent pay. This analytical result applies to the filter's steady state without recurring payments and with known, constant variances. I will relax these assumptions in the quantitative model, where workers learn jointly about permanent pay and future risk. As all terms in Proposition 1 can then change, I am able to assess how large the effect is and whether bonus recognition can offset the greater earnings variation at higher incomes.

Risk Updating and Persistence. Recognising temporary payments may also affect how workers revise their assessment of earnings risk. For instance, a worker who cannot identify temporary payments may need more evidence to distinguish a change in permanent-pay risk from temporary noise. Recognising temporary payments may therefore help her correct earlier errors in her assessment of risk more quickly.

I use the squared pay surprise, $(e^j_t)^2$, as a noisy reading of risk, since large surprises of either sign can indicate greater variance. To obtain a Kalman update, I approximate the ARCH risk process and the signal constructed from this surprise by a linear Gaussian system. As before, I treat the parameters as known and evaluate the filters in steady state.

Here, h_t approximates permanent-pay risk q_t, and s^j_t is the worker's noisy reading of this risk. The approximation preserves the ARCH conditional mean, with persistence $0 < \alpha < 1$ and risk-innovation variance $Q > 0$. The signal has mean-zero noise with some variance Ω_j, which I keep constant in the analytical comparison, and where a smaller Ω_j allows the worker to assess risk more precisely.[8]

Proposition 4 (Pay information and the correction of risk assessments)**.** *With known parameters, no recurring payments and steady-state filtering in the Gaussian approximation, the worker updates her assessment of risk according to*

$$\begin{aligned} \widehat{h}^j_{t|t-1} &= (1-\alpha)\bar{q} + \alpha \widehat{h}^j_{t-1}, \\ \widehat{h}^j_t &= \widehat{h}^j_{t|t-1} + K^j_h\big(s^j_t - \widehat{h}^j_{t|t-1}\big), \end{aligned} \tag{17}$$

where $\widehat{h}^j_t$ is the worker's estimate of current risk and K^j_h is the steady-state Kalman gain. Positive variance in the separately observed component, $\sigma_b^2 > 0$, implies a more precise risk signal, $\Omega_1 < \Omega_0$, and a larger gain, $K^1_h > K^0_h$. Holding $\bar{q}$, α, Q and total temporary variance fixed, a larger separately observed share of temporary variance raises the gain under $\mathcal{I}^1_t$. Appendix I.4 derives the gain and the error recursion.

Recognising temporary payments removes their variation from the pay signal and reduces uncer-

[8] In Appendix I.4, I derive Ω_j from the variance of the squared pay surprise, using the pay filter in Proposition 3 and evaluating risk at its common long-run level $\bar{q}$. The approximation treats state innovations and signal errors as mutually independent, independent across months and independent of the Gaussian initial state.

tainty about previous permanent pay, making new payments more informative about risk. The worker therefore places greater weight on new evidence and corrects earlier errors more quickly. An earlier estimation error carries forward with coefficient $\alpha(1 - K_h^j)$, so a larger gain reduces its influence on subsequent assessments. Whether this also lowers the persistence of reported earnings risk depends on how the other components of forecast uncertainty outlined in Proposition 1 evolve, which I address in the next section.

5 Quantitative Analysis

I now examine the generalised pay-composition mechanism in Section 4 with unknown variance parameters, time-varying risk and scheduled payments. In this exercise I assess how much payment information can explain income differences in the level and persistence of perceived earnings risk. I externally calibrate the earnings process to the SIPP and hold the number of payments observed constant across income groups. With information about bonus amounts and payment dates, the model reproduces part of the SCE persistence gap and generates a modest decline in mean perceived risk with income. Given this calibration, I then examine how bonus recognition changes responses to temporary payments and discuss a policy application comparing how annual transfers to households and monthly instalments affect perceived risk.

5.1 Quantitative Environment

I simulate monthly earnings histories for workers who remain in the same job. Earnings follow the process in Section 4.1, including the recurring payments:

$$\begin{aligned} w_t &= p_t + u_t + b_t, \qquad p_t = p_{t-1} + \psi_t, \\ \psi_t &= \sqrt{q_t}\,\varepsilon_t, \qquad\quad q_{t+1} = (1-\alpha)\bar{q} + \alpha\psi_t^2, \\ b_t &= v_t + D_t a_t. \end{aligned}$$

The shocks are independent and centred Gaussian, with $\varepsilon_t \sim N(0,1)$ and temporary-payment variances σ_u^2, σ_v^2 and σ_a^2. A share ρ of workers receives an annual payment, with $D_t = 1$ in their scheduled month and zero otherwise. I draw each recipient's payment month uniformly from the twelve calendar months and keep it fixed across years.

I compare each worker's risk assessments under the three information sets, holding her earnings history fixed. She receives the same annual payments in each case, but knows their schedule only under $\mathcal{I}_t^2$. Under $\mathcal{I}_t^0$ and $\mathcal{I}_t^1$, I assume that she does not learn this schedule from past receipts. After each payment, she updates her beliefs jointly about permanent pay, its current risk and the parameters governing her earnings.[9] I then use these beliefs to calculate her twelve-month earnings forecast variance, giving a monthly series of perceived risk under each information set that I can compare with the levels and persistence observed in the SCE.

To move from the fixed worker types used in the estimation towards a more flexible heterogeneous-agent model, I allow workers to move across the income distribution. I let temporary-pay parameters vary smoothly with permanent income, so the composition of their payments changes with their position in the distribution. For instance, as workers' permanent income rises, a larger share of

[9] I explain the numerical implementation of the joint update in Appendix J.2, where I track possible permanent-pay histories and parameter values and revise their probabilities after each payment. Learning about annual-payment variance uses a separate approximation.

temporary earnings variation comes from bonuses they can distinguish from regular pay.[10]

5.2 Simulation and Calibration

I calibrate the earnings process to the SIPP and specify common initial beliefs across income groups, then simulate workers' risk reports using the same number of payments observed across groups and compare their persistence with the SCE.

Earnings Calibration. I use the SIPP evidence in Sections 2 and 3 to calibrate the earnings process workers face. For each income group, I use the variance decomposition in the earnings benchmark to determine long-run permanent variance $\bar{q}$ and total temporary variance, then the component decomposition to separate the bonus contribution from other temporary variance σ_u^2. These targets reflect the approximately flat permanent variance and larger bonus contribution at higher incomes documented in Finding 1. I use the recurrence evidence in Section 3.3 to determine the share of workers receiving annual payments, ρ, and split the measured bonus contribution between annual and irregular payments. To calibrate ARCH persistence α, I match persistence in the SIPP volatility proxy to the same statistic computed from simulated earnings, obtaining $\alpha = 0.44$.[11] Table 7 reports the earnings parameters, prior assumptions and number of payments observed.

Initial Beliefs. Workers begin with common priors over the variance parameters, so income differences in their beliefs emerge from their own payments rather than income-specific starting values. I set the prior means of long-run permanent and total monthly temporary variance to their SIPP cross-group averages, 8.65 and 3.06. I assume lognormal priors with a common log standard deviation of 2.5, keeping the variances positive and leaving workers with substantial initial uncertainty about earnings risk.

The prior links beliefs about bonus and other temporary variance so that workers can learn about overall temporary-pay variability from the bonuses they observe. Under $\mathcal{I}^0$ and $\mathcal{I}^1$, an uncertain share divides total temporary variance between the two components. A large observed bonus can therefore suggest greater temporary variation overall, leading the worker to revise her assessment of other temporary variance as well. Under $\mathcal{I}^2$, I apply this construction to $\sigma_u^2 + \sigma_v^2$ and specify a separate prior for annual-payment variance σ_a^2. Workers also learn ARCH persistence α, starting from $U(0, 0.55)$.[12]

5.3 Model Fit

I assess the model's fit to persistence across income groups and to their mean perceived risk ratios. The simulation follows 8,192 workers per income group from job entry as they learn from successive payments. To guide the choice of the number of payments observed, A, I compare persistence under $\mathcal{I}^2$ with the SCE estimates (Figure 4), using the same system GMM specification on twelve monthly risk assessments starting at payment A.

I use persistence to guide the choice of A because it responds more strongly to accumulated

[10] The simulated temporary variances change with permanent income according to the schedules in Appendix J.4. The worker estimates constant variance parameters and revises her beliefs after each payment, which approximates learning about this changing process. Long-run permanent variance and annual-payment eligibility remain fixed at job entry.

[11] The persistence of the SIPP volatility proxy need not equal α, since temporary payments also contribute to the proxy. I therefore simulate earnings at different values of α and compare the resulting persistence estimates with the SIPP using the same proxy and estimator. Appendix J.3 reports the calibration.

[12] I set the upper bound at 0.55 to keep the long-run variance of current risk finite. Under the Gaussian ARCH process, this requires $\alpha < 1/\sqrt{3} \approx 0.577$.

Table 7: Calibration and assumptions

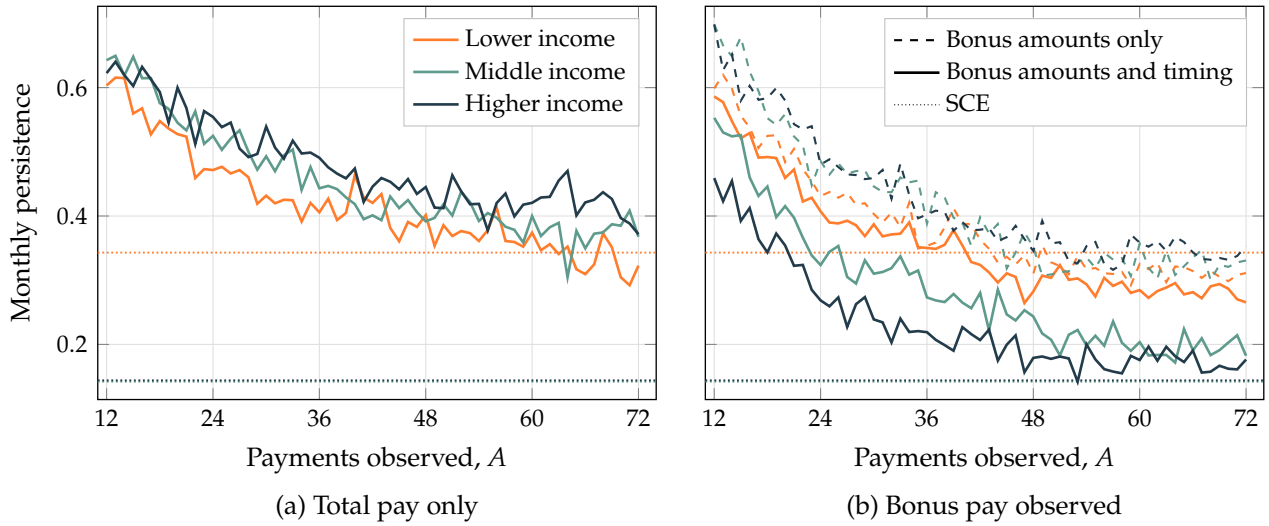
Parameter	Lower	Middle	Higher	Basis
A. Earnings process				
Long-run permanent variance, $\bar{q}$	8.93	8.64	8.39	SIPP moments
Other temporary variance, σ_u^2	1.32	1.44	1.32	SIPP moments
Average bonus variance, σ_b^2	0.73	1.62	2.74	SIPP moments
ARCH persistence, α		0.44		SIPP proxy
B. Annual bonus timing				
Share with annual payments, ρ	0.070	0.136	0.194	SIPP timing
Annual-payment variance, σ_a^2	21.9	47.0	53.1	SIPP timing
Irregular-bonus variance, σ_v^2	0.60	1.09	1.88	SIPP timing
C. Common prior beliefs				
Mean permanent variance, $\widetilde{\bar{q}}$		8.65		SIPP average
Mean monthly temporary variance, $\widetilde{\sigma}_T^2$		3.06		SIPP average
Log standard deviations, τ_q, τ_T		2.5, 2.5		Assumed
Bonus share of monthly temporary variance		$U(0,1)$		Assumed
ARCH persistence, $\widetilde{\alpha}$		$U(0, 0.55)$		Assumed
Mean annual-payment variance, $\widetilde{\sigma}_a^2$		40.7		SIPP average
Annual-variance log standard deviation		1.5		Assumed
D. Payments observed				
Payments observed, A		48		Calibrated

Notes: Variances are in squared 100-log-point units. Income-specific entries are calibration targets for group averages at job entry.

payment information than mean-risk ratios, which change little as workers observe more payments. These ratios remain untargeted, providing a separate test of whether measured pay composition and workers' information can explain the SCE income gradient. Appendix Table J.2 reports the full benchmark results under all three information sets.

Persistence of Perceived Earnings Risk. Figure 4 shows that total-pay information, $\mathcal{I}^0$ in panel (a), does not generate systematically lower persistence at higher incomes. The reversal becomes clear in panel (b) when workers know payment dates: persistence under $\mathcal{I}^2$, shown by the solid lines, declines with income. Observing bonus amounts alone, $\mathcal{I}^1$, lowers persistence but does not reproduce this ordering. The distinction therefore concerns what workers know about future payments, beyond recognising the bonus they have just received.

The decline in persistence with income reflects how workers use information about the recurring bonuses conjectured in Finding 3. To understand this difference, consider a worker receiving a scheduled annual bonus, as illustrated in Appendix Figure J.1. Under $\mathcal{I}^1$, she recognises the receipt as temporary, but a large bonus can lead her to expect greater variation in future bonus amounts. This revised estimate enters successive forecasts until further payments change her assessment, so perceived risk can remain elevated even after earnings return to their previous level. However, under $\mathcal{I}^2$, she can assign annual-payment uncertainty to the scheduled month. A forecast made in that month includes uncertainty about next year's bonus; the following month's forecast concerns ordinary pay, so that contribution disappears even if her assessment of bonus variability is unchanged. Annual payments are more common and more variable at higher incomes, making these temporary increases a larger source of variation in their perceived risk. Knowing payment dates therefore reduces how



(a) Total pay only

(b) Bonus pay observed

Figure 4: Persistence of perceived earnings risk by number of payments observed

Notes: Each point is an estimate of monthly perceived-risk persistence from assessments after payments A to $A + 11$; adjacent points share eleven assessments. Income groups are reassigned at each A using earnings over payments $A - 11$ to A. Dotted lines show the SCE estimates.

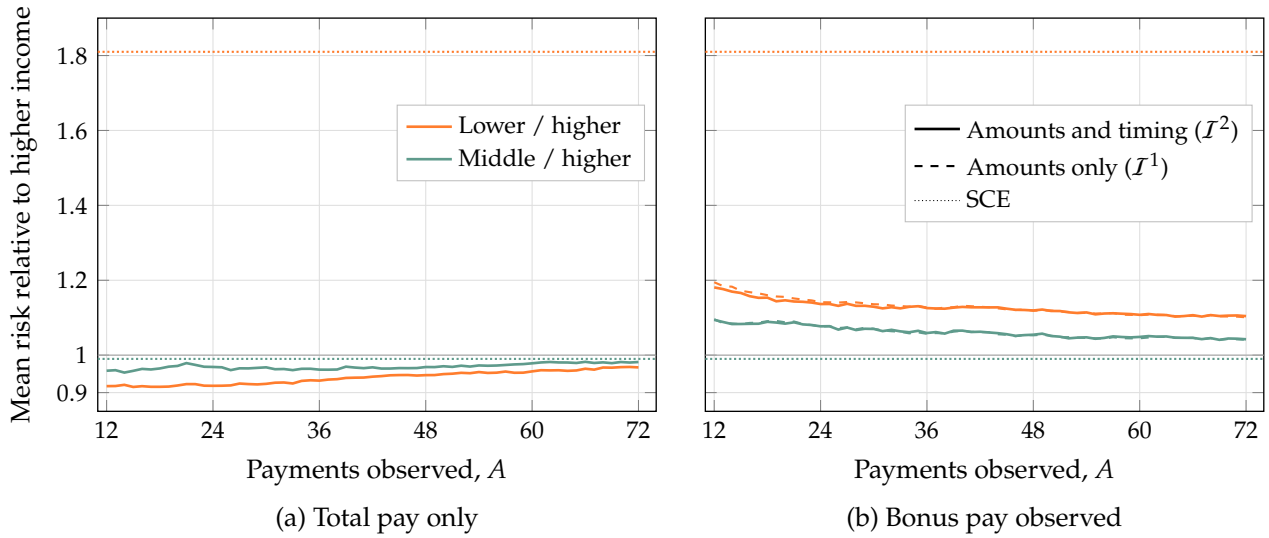
strongly one month's risk assessment carries into the next, particularly at higher incomes, without requiring faster learning about the underlying earnings process.

The resulting lower–higher persistence gap under $\mathcal{I}^2$ ranges from 0.076 to 0.188 when the number of payments observed ranges from 12 to 72, compared with 0.199 in the SCE. Payment timing thus reverses the income ordering under $\mathcal{I}^0$ and generates part of the observed gradient, although the fit remains imperfect: at $A = 48$, persistence is too high at middle and higher incomes and too low at lower incomes. Since persistence under $\mathcal{I}^2$ changes little after the first 48 payments, I retain $A = 48$ to limit the computational cost of simulating additional payments.

The Level of Perceived Earnings Risk. Figure 5 compares mean perceived risk across income groups relative to higher-income workers. Under total-pay information, $\mathcal{I}^0$ in panel (a), both ratios lie below one: higher-income workers perceive the most risk. Observing bonus amounts reverses this ordering in panel (b), where lower-income workers perceive more risk than middle- and higher-income workers. The lines under $\mathcal{I}^1$ and $\mathcal{I}^2$ nearly coincide. Recognising bonus amounts therefore matters for the income gradient in mean perceived risk, and knowing payment dates adds little to this effect.

The reversal reflects how workers interpret the earnings variation documented in Finding 1. Higher-income workers experience more bonus variation, and with only total pay observed, their estimates attribute part of it to permanent risk. Observing bonuses separately reduces their estimates of permanent risk more strongly. Workers remain uncertain about future bonus amounts, but expect less variation from permanent shocks, which accumulate over the twelve-month forecast horizon. This inference also depends on the prior over temporary pay composition: under the maintained prior, a large bonus can raise the worker's assessment of other temporary variance, leaving less of the variation in pay net of bonuses attributed to permanent shocks. Bonus recognition thus affects beliefs about future permanent risk as well as uncertainty about current permanent pay. This reduction in perceived risk from observing bonuses is consistent with Wang (2026), who finds that conventional risk estimates partly capture earnings variation workers already anticipate.

The resulting income differences nevertheless remain modest. At $A = 48$, observing bonus amounts



(a) Total pay only

(b) Bonus pay observed

Figure 5: Relative perceived earnings risk by number of payments observed

Notes: Each point is a ratio of group mean perceived risk immediately after payment A, with higher-income workers as the denominator. Income groups are reassigned at each A using earnings over payments $A - 11$ to A. Dotted lines show the SCE ratios of 1.81 and 0.99; the thin grey line indicates equal mean risk. Appendix Figure J.2 shows the same plot but with corresponding levels.

raises the lower–higher mean-risk ratio from 0.95 to 1.11, compared with 1.81 in the SCE; the middle–higher ratio rises from 0.97 to 1.05, compared with 0.99 (see Appendix Table J.2 for the full results). The difficulty in matching the mean-risk ratios is that expected permanent risk dominates the twelve-month forecast, and the calibrated permanent variances are similar across income groups. Workers' estimates remain relatively close as workers observe more payments, so bonus recognition generates only modest income differences and changing A provides little potential to close the remaining gap.[13]

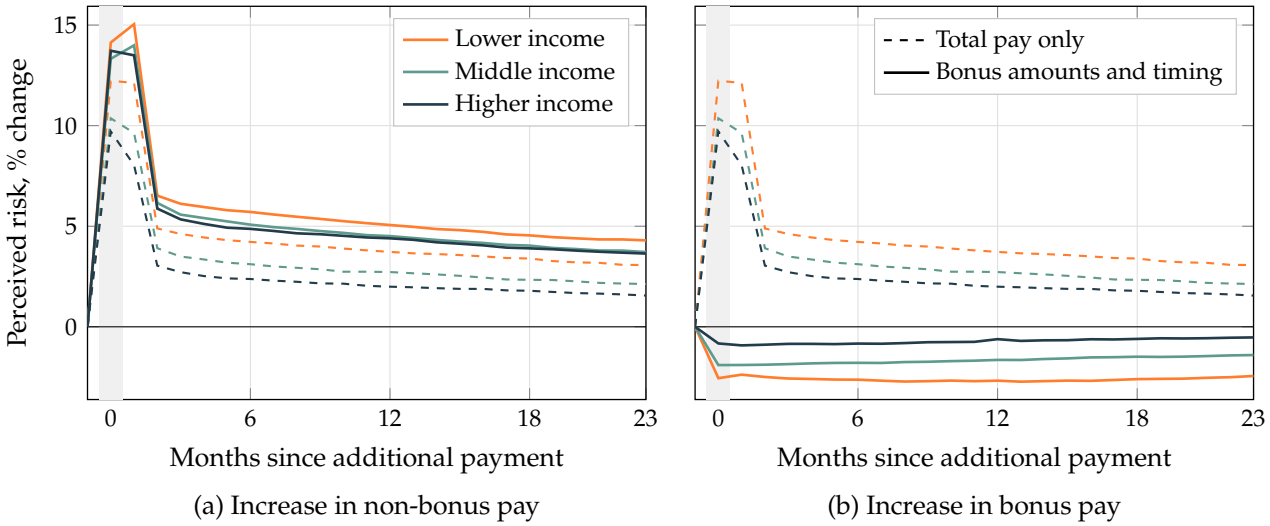
5.4 Model Applications

I now use the calibrated model to examine the implications of these findings in two practical settings. I first study how bonus recognition changes perceived earnings risk after a temporary pay increase, then consider a policy example to assess how transfers should be delivered to limit increases in perceived earnings risk.

Responses to Temporary Payments. I consider a worker whose pay temporarily rises and examine what she learns from this increase about her future earnings risk. When its source is unclear, the payment may lead her to expect greater variation in permanent pay, whilst recognising it as a bonus allows her to attribute the increase and its subsequent disappearance to temporary pay instead. I reproduce this situation in two separate cases, adding a one-month increase of five log points to non-bonus pay u_t in panel (a) and to bonus pay b_t in panel (b). Figure 6 compares the resulting changes in perceived risk under $\mathcal{I}^0$ and $\mathcal{I}^2$, relative to the same earnings histories without the additional payment.

Under $\mathcal{I}^0$, the dashed responses coincide across panels: perceived risk rises by 10–12 percent and remains elevated after the payment ends. The worker interprets the rise and subsequent fall as

[13] A stronger reduction in perceived permanent risk could arise if workers subtract identified temporary payments but continue assigning the same share of the remaining variation to permanent shocks. The present model does not impose this restriction. However, there is evidence of insufficient sensitivity to signal strength (Augenblick, Lazarus and Thaler, 2025) and cognitive uncertainty (Enke and Graeber, 2023) that motivates adding it.



(a) Increase in non-bonus pay

(b) Increase in bonus pay

Figure 6: Perceived risk after a temporary increase in pay

Notes: Simulation of 24,576 workers receiving a one-month increase of five log points after 48 payments (shaded month). Lines show percentage changes in mean twelve-month forecast variance relative to the same workers' histories without the increase. Income groups are fixed using earnings over the preceding twelve payments.

evidence of greater permanent earnings variability, so returning to her previous pay does not restore her previous risk assessment. Under $\mathcal{I}^2$, observing ordinary bonuses separately does not resolve this uncertainty when the additional payment enters non-bonus pay. When the additional payment itself is identified as a bonus, however, subtracting it leaves the signal of permanent pay unchanged and the large positive response disappears. The small decline in perceived risk follows the inference about pay composition discussed in the comparison of mean perceived risk: under the maintained prior, a larger bonus can raise perceived temporary variability and reduce the variation attributed to permanent shocks. Because permanent shocks accumulate over the forecast horizon, this reduction outweighs the increase in temporary-pay risk in this calibration. The experiment therefore suggests that clearly identifying the source and temporary nature of a payment can limit its effect on perceived earnings risk without changing the amount workers receive.

Policy Application. Now in a policy application, consider a government transfer intended to stimulate the economy, such as a stimulus payment during the Covid pandemic. The previous experiment illustrated how a one-off receipt can affect perceived earnings risk when it is not distinguished from ordinary pay. I now extend this exercise to a longer-term recurring transfer, for instance an annual tax rebate. Using the calibrated model, I examine whether paying it all at once or spreading it equally across twelve months produces smaller changes in perceived earnings risk, and how recognising the transfer affects this comparison.

Each worker receives a fixed annual amount equal to 6 percent of her average monthly earnings over the preceding year. I pay this amount once a year or divide it into twelve equal instalments, keep the programme in place for three years, and follow beliefs for another year after it ends. For each delivery schedule, both learners observe ordinary bonus amounts and dates under $\mathcal{I}^2$. One also identifies each government transfer and its amount when received; the other sees it only as part of total receipts.

Figure 7 shows a large difference between the two delivery schedules when transfers are not identified (solid lines). The first annual payment in panel (a) raises mean perceived risk by 19–22

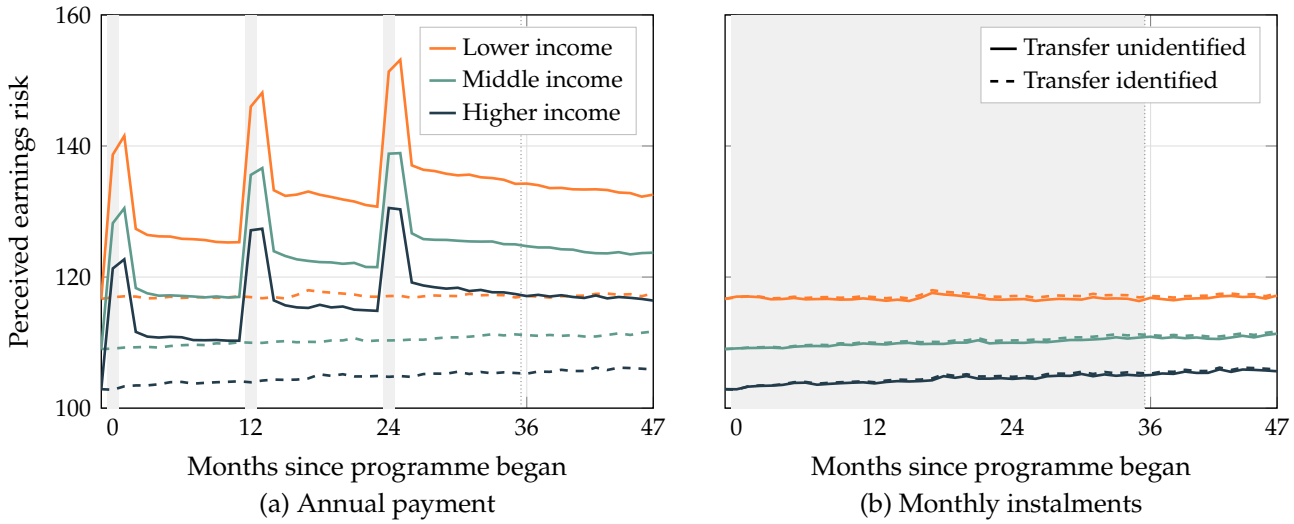


Figure 7: Transfer delivery and perceived earnings risk

Notes: Simulation of 24,576 workers. Lines show mean twelve-month earnings forecast variance in squared 100-log-point units. Both learners observe ordinary bonus amounts and dates ($\mathcal{I}^2$). Dashed lines identify the transfer at receipt and coincide with the no-transfer paths. Income groups update monthly using earnings over the preceding twelve payments, excluding transfers. Shading marks receipt; the dotted line marks programme end.

variance units and later receipts produce further increases. The worker reads the large receipt and the subsequent return to ordinary pay as evidence of greater earnings volatility. Monthly instalments in panel (b) produce much smaller revisions, with declines of less than one variance unit during the programme and no comparable spike at withdrawal. A fixed cash instalment also dampens proportional fluctuations in observed pay, consistent with these small negative responses.

The annual-payment effect persists after the programme ends. At the end of the follow-up year, perceived risk remains about 11–15 variance units above its no-transfer path, with the remaining response concentrated in beliefs about future permanent shocks. Recognising the transfer gives the worker an explanation for the movements in her receipts. With each receipt identified, she subtracts the government payment before learning about earnings, so perceived earnings risk follows the no-transfer path under either delivery schedule (dashed lines). These results favour clearly identifying the source and amount of government transfers, so recipients can distinguish support from changes in their earnings. Avoiding a rise in perceived earnings risk may be particularly important when the aim is to stimulate spending among households with limited liquid assets, who tend to spend a larger share of additional income (Fagereng, Holm and Natvik, 2021). Where recipients cannot distinguish transfers from earnings, monthly instalments limit the increase in perceived risk without reducing the annual amount they receive.

Macroeconomic Implications. I conclude the quantitative analysis by considering what these findings imply for macroeconomic consumption–saving models, focusing on precautionary motives, calibration and transfer design.

First, in heterogeneous-agent models with incomplete markets, households save to protect consumption against earnings losses they cannot fully insure (Aiyagari, 1994; Carroll, 1997). The desired buffer depends on the risk households perceive, so information about pay composition can affect both its size and its distribution across incomes. My model generates part of the income gradient in perceived risk from common initial beliefs, as workers learn from payments with different compositions.

This provides an economic basis for heterogeneity in beliefs, which can help consumption models explain differences in wealth and spending (Carroll et al., 2017). Saving motives can then evolve with workers' payment histories. Greater perceived risk need not imply greater observed saving, however, since households may lack the resources to accumulate the protection they desire.

Second, learning can sustain a precautionary motive after the payment that prompted it has disappeared. In the temporary-payment experiment, an unidentified receipt raises the worker's assessment of permanent earnings variability, and returning to ordinary pay does not immediately reverse that assessment. For a given expected earnings path, the resulting desire to protect future consumption could therefore outlast the additional income. This connects to the precautionary saving generated by uncertainty about individual income components in Wang (2004). Predictable calendar changes in forecast risk need not involve such revisions to beliefs, meaning that a consumption model must therefore distinguish lasting changes in perceived earnings variability from changes in when uncertain payments occur.

Third, calibrating households' perceived risk directly from measured earnings variation can overstate the uncertainty they face. SIPP estimates used in models of precautionary saving and portfolio choice (Bayer et al., 2019; Choukhmane and de Silva, 2026), but workers may anticipate some variation that the econometrician treats as risk. Assigning that variation to their uncertainty can generate excessive precautionary buffers, contributing to the difficulty some consumption models face in matching low liquid wealth and large spending responses to additional income. Tao Wang (2026) shows that incorporating subjective expectations and risks improves both predictions. My analysis instead provides the micro-founded mechanism through which payment information generates a wedge between measured and perceived risk.

Lastly, transfer delivery can affect how households divide additional income between consumption and saving. An unlabelled transfer relaxes the household's budget constraint, but may also lead it to expect less stable earnings. The household may then save more of the additional income to protect future consumption. Monthly instalments instead induce smaller revisions to perceived risk, suggesting less scope for this precautionary response than an annual payment of the same entitlement. This provides one mechanism through which delivery could affect spending, although other effects may work in the opposite direction. Sahm et al. (2012), for instance, find a higher reported propensity to spend lump-sum payments than reduced withholding. My results therefore suggest that monthly delivery can limit revisions to perceived earnings risk without necessarily producing the largest spending response. That response also depends on households' liquidity needs, borrowing constraints and understanding of the programme. Quantifying these effects requires a consumption–saving model in which households respond to both the additional income and their revised beliefs, which I leave as an avenue for future research.

6 Conclusion

In this paper I show how information about pay composition can explain part of the difference between measured and perceived earnings risk across incomes. Empirically, I show that bonuses account for more temporary earnings variation and show stronger annual recurrence at higher incomes. In a learning model calibrated to these patterns, recognising bonuses generates a modest decline in mean perceived risk with income. Knowing payment dates helps reproduce the persistence gradient because annual-payment uncertainty enters and leaves successive forecasts, without requiring faster learning

at higher incomes. The model nevertheless leaves much of the difference in mean risk unexplained. Transfer experiments show that unrecognised annual payments can raise perceived risk beyond the month of receipt, with much smaller effects when the same entitlement is paid monthly. These findings therefore suggest that models of consumption and saving should account for what households know about their payments when translating observed earnings variation into the risk they perceive, with potentially important quantitative implications for asset accumulation and spending responses.

References

Aiyagari, S. Rao (1994). Uninsured Idiosyncratic Risk and Aggregate Saving. *Quarterly Journal of Economics*, 109(3), 659–684.

Augenblick, Ned, Eben Lazarus and Michael Thaler (2025). Overinference from Weak Signals and Underinference from Strong Signals. *Quarterly Journal of Economics*, 140(1), 335–401.

Bayer, Christian, Ralph Lütticke, Lien Pham-Dao and Volker Tjaden (2019). Precautionary Savings, Illiquid Assets, and the Aggregate Consequences of Shocks to Household Income Risk. *Econometrica*, 87(1), 255–290.

Ben-David, Itzhak, Elyas Fermand, Camelia M. Kuhnen and Geng Li (2018). Expectations Uncertainty and Household Economic Behavior. NBER Working Paper No. 25336.

Blundell, Richard and Stephen Bond (1998). Initial Conditions and Moment Restrictions in Dynamic Panel Data Models. *Journal of Econometrics*, 87(1), 115–143.

Bound, John and Alan B. Krueger (1991). The Extent of Measurement Error in Longitudinal Earnings Data: Do Two Wrongs Make a Right? *Journal of Labor Economics*, 9(1), 1–24.

Caplin, Andrew, Victoria Gregory, Eungik Lee, Søren Leth-Petersen and Johan Sæverud (2026). Subjective Earnings Risk. *Review of Economic Studies*, advance online publication, rdag070.

Carroll, Christopher D. (1997). Buffer-Stock Saving and the Life Cycle/Permanent Income Hypothesis. *Quarterly Journal of Economics*, 112(1), 1–55.

Carroll, Christopher, Jiri Slacalek, Kiichi Tokuoka and Matthew N. White (2017). The Distribution of Wealth and the Marginal Propensity to Consume. *Quantitative Economics*, 8(3), 977–1020.

Choukhmane, Taha and Tim de Silva (2026). What Drives Investors' Portfolio Choices? Separating Risk Preferences from Frictions. *Journal of Finance*, 81(1), 5–48.

Di Maggio, Marco, Amir Kermani, Rodney Ramcharan, Vincent Yao and Edison Yu (2022). The Pass-Through of Uncertainty Shocks to Households. *Journal of Financial Economics*, 145(1), 85–104.

Druedahl, Jeppe and Thomas Høgholm Jørgensen (2020). Can Consumers Distinguish Persistent from Transitory Income Shocks? *Economic Journal*, 130(632), 2410–2437.

Engelberg, Joseph, Charles F. Manski and Jared Williams (2009). Comparing the Point Predictions and Subjective Probability Distributions of Professional Forecasters. *Journal of Business & Economic Statistics*, 27(1), 30–41.

Enke, Benjamin and Thomas Graeber (2023). Cognitive Uncertainty. *Quarterly Journal of Economics*, 138(4), 2021–2067.

Fagereng, Andreas, Martin B. Holm and Gisle J. Natvik (2021). MPC Heterogeneity and Household Balance Sheets. *American Economic Journal: Macroeconomics*, 13(4), 1–54.

Ganong, Peter, Pascal J. Noel, Christina Patterson, Joseph S. Vavra and Alexander Weinberg (2025). Earnings Instability. NBER Working Paper No. 34227, revised July 2026.

Grigsby, John, Erik Hurst and Ahu Yildirmaz (2021). Aggregate Nominal Wage Adjustments: New Evidence from Administrative Payroll Data. *American Economic Review*, 111(2), 428–471.

Guvenen, Fatih, Fatih Karahan, Serdar Ozkan and Jae Song (2021). What Do Data on Millions of U.S. Workers Reveal About Lifecycle Earnings Dynamics? *Econometrica*, 89(5), 2303–2339.

Kim, Gwangmin and Carola Binder (2023). Learning-through-Survey in Inflation Expectations. *American Economic Journal: Macroeconomics*, 15(2), 254–278.

Krüger, Fabian and Lora Pavlova (2024). Quantifying Subjective Uncertainty in Survey Expectations. *International Journal of Forecasting*, 40(2), 796–810.

Lemieux, Thomas, W. Bentley MacLeod and Daniel Parent (2009). Performance Pay and Wage Inequality. *Quarterly Journal of Economics*, 124(1), 1–49.

Ma, Yueran, Tiziano Ropele, David Sraer and David Thesmar (2020). A Quantitative Analysis of Distortions in Managerial Forecasts. NBER Working Paper No. 26830.

Meghir, Costas and Luigi Pistaferri (2004). Income Variance Dynamics and Heterogeneity. *Econometrica*, 72(1), 1–32.

Orlik, Anna and Laura Veldkamp (2024). Understanding Uncertainty Shocks and the Role of Black Swans. *Journal of Economic Theory*, 222, 105905.

Pistaferri, Luigi (2001). Superior Information, Income Shocks, and the Permanent Income Hypothesis. *Review of Economics and Statistics*, 83(3), 465–476.

Sahm, Claudia R., Matthew D. Shapiro and Joel Slemrod (2012). Check in the Mail or More in the Paycheck: Does the Effectiveness of Fiscal Stimulus Depend on How It Is Delivered? *American Economic Journal: Economic Policy*, 4(3), 216–250.

Sockin, Jason and Michael Sockin (2025). Variable Pay and Risk Sharing Between Firms and Workers. IZA Discussion Paper No. 17644.

Thakral, Neil and Linh T. Tô (2025). Excess Anticipation-Dependence in Consumption. Working paper, February.

Wang, Neng (2004). Precautionary Saving and Partially Observed Income. *Journal of Monetary Economics*, 51(8), 1645–1681.

Wang, Tao (2026). Perceived versus Calibrated Income Risks in Heterogeneous-Agent Consumption Models. Working paper, September.

A Measuring Perceived Earnings Risk

A.1 Survey Question and Variance Construction

The SCE question on earnings uncertainty (Q24) asks:

> *"Suppose again that, 12 months from now, you are working in the exact same job at the same place you currently work, and working the exact same number of hours. In your view, what would you say is the percent chance that 12 months from now..."*

The response options complete the statement *"Your earnings on this job, before taxes and deductions, will have..."* and cover increases and decreases of 0–2%, 2–4%, 4–8%, 8–12%, and 12% or more. For respondents with multiple jobs, the question inserts "main" before "job". I calculate perceived earnings risk as the variance of the discrete distribution obtained by assigning each bin's probability to its midpoint. I use ±1, ±3, ±6 and ±10 percent for the closed intervals and ±16 percent for the open-ended tails, following the medium-width specification of Krüger and Pavlova (2024).

A.2 Adjustment and Sample Comparisons

To account for differences in observed characteristics and interview dates across income groups, I regress the bin-midpoint variance on age, age squared, sex, education, state, calendar year-month and interview-position indicators, using survey weights. Interview position counts consecutive monthly reports within a response spell, with positions twelve and above grouped together. I add the weighted mean variance in the regression sample back to each residual to obtain the adjusted measure. I use this same measure in the subsamples below, so differences in their reported means come from which respondents are retained; I do not re-estimate the adjustment.

Adjusting for these characteristics reduces the lower–higher gap in perceived variance (Table A.1). The adjusted gap is close to zero among high-numeracy respondents and remains large among low-numeracy respondents, who supply 47.3, 25.9 and 10.6 percent of weighted reports in the lower-, middle- and higher-income groups. Restricting the sample to high-numeracy respondents retains 28,738 reports from 3,021 respondents, compared with 35,697 from 3,790 overall. Because this also changes the composition of the income groups, the comparison cannot separate differences in uncertainty from differences in reporting. Appendix D.2 examines persistence in this subsample.

Table A.1: Perceived variance and numeracy

Measure and sample	Lower	Middle	Higher	Lower–higher difference (SE)
Raw, all respondents	22.85	11.37	8.74	14.11 (1.49)
Adjusted, all respondents	20.62	11.33	11.41	9.22 (1.38)
Raw, high numeracy	11.71	8.90	8.28	3.43 (1.25)
Adjusted, high numeracy	10.95	9.41	11.15	−0.21 (1.13)
Raw, low numeracy	35.28	18.43	12.65	22.63 (3.05)
Adjusted, low numeracy	31.41	16.83	13.52	17.89 (2.87)

Notes: Survey-weighted means in squared percentage points; respondent-clustered standard errors of differences in parentheses. Numeracy uses the first available SCE classification.

A.3 Income Cut-offs

I check whether the lower–higher gap depends on the chosen income boundaries by varying them while holding the reports, survey weights and adjustment fixed (Table A.2). The adjusted gap remains positive when the lower group is below \$20,000 or the higher group above \$200,000, but falls when the lower group includes all households below \$100,000. The largest differences arise when the comparison isolates the lowest-income households.

Table A.2: Perceived variance under alternative income cut-offs

Household income		Adjusted mean		Difference (SE)	p
Lower group	Higher group	Lower	Higher		
$< \$50,000$	$\geq \$100,000$	20.62	11.41	9.22 (1.38)	< 0.001
$< \$20,000$	$\geq \$100,000$	37.49	11.41	26.09 (3.74)	< 0.001
$< \$50,000$	$\geq \$200,000$	20.62	13.95	6.67 (1.54)	< 0.001
$< \$20,000$	$\geq \$200,000$	37.49	13.95	23.54 (3.81)	< 0.001
$< \$100,000$	$\geq \$200,000$	15.28	13.95	1.32 (1.05)	0.207

Notes: Survey-weighted adjusted means in squared percentage points; respondent-clustered standard errors in parentheses. Differences are lower minus higher. Cut-offs follow SCE bands; the lowest contains 102 respondents.

B The Distribution of Pay Components

The distribution of positive bonuses helps explain why their mean can substantially exceed the payment a typical recipient receives (Table B.1). At higher incomes, the median is \$2,030 and the ninety-ninth percentile is \$168,000, compared with the mean of \$10,150 in Table 1. In months when they are received, bonuses account for an average of 21.6–30.4 percent of monthly pay across income groups.

C The Earnings Benchmark

C.1 Hours and Sample Period

I assign workers to income groups using their median annualised household income and retain same-job earnings changes whether hours change or not. To assess the role of hours, I also require equal weekly hours at both endpoints of each change and across all three months used for an adjacent pair. This restriction retains 414,575, 690,502 and 869,519 changes in the lower-, middle- and higher-income groups and excludes only one worker, but removes months that contribute disproportionately to squared earnings changes. Figure C.1 shows how hours variation differs across income groups.

Requiring unchanged hours lowers estimated permanent variation and changes its income pattern from slightly declining to rising, while temporary variation rises with income under either rule (Table C.1). Restricting the unchanged-hours sample to 2017–2023 raises both components. For each period, I re-estimate the earnings regression and pooled trimming thresholds, then hold these fixed when comparing the two hours restrictions within that period.

Table B.1: Monthly receipt and the distribution of positive payments

	Receipt (%)	Median ($)	90th percentile ($)	99th percentile ($)	Share of pay when received (%)
A. Lower income					
Bonuses	0.77	300	2,000	13,000	22.5
Commissions	2.48	1,100	3,200	14,700	76.5
Tips	2.56	400	1,500	4,180	38.5
Overtime	2.65	225	900	2,400	12.1
B. Middle income					
Bonuses	1.22	500	5,000	20,000	21.6
Commissions	2.90	1,800	5,000	14,000	67.1
Tips	1.60	540	2,000	6,230	36.7
Overtime	4.12	400	1,500	5,980	13.8
C. Higher income					
Bonuses	1.89	2,030	21,000	168,000	30.4
Commissions	3.45	3,220	14,300	59,500	59.6
Tips	1.12	500	2,800	8,540	35.3
Overtime	3.67	560	2,900	9,350	14.8

Notes: Survey-weighted SIPP estimates, 2013–2023; nominal dollars. Quantiles and mean pay shares condition on receipt. Hours may change.

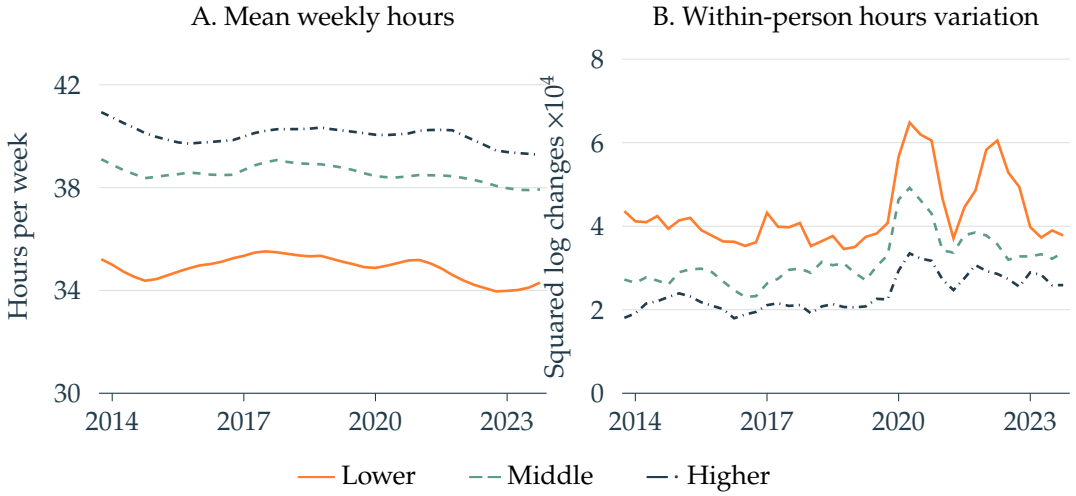


Figure C.1: Hours worked and hours variation by income group

Notes: Survey-weighted SIPP means, 2013–2023, within jobs: weekly hours (A) and squared monthly log-hours changes scaled by 10^4 (B). January changes are excluded; squared changes are trimmed at pooled 1st/99th percentiles. Lines are four-quarter trailing averages.

C.2 Dependence beyond One Month

The benchmark imposes restrictions beyond the two moments used to recover $\mathcal{P}_g$ and $\mathcal{T}_g$. For mean-zero earnings changes, it predicts

$$\mathbb{E}[d_{it}d_{i,t-k} \mid g] = \begin{cases} -\mathcal{T}_g, & k = 1, \\ 0, & k \geq 2. \end{cases} \tag{C.1}$$

Table C.1: Earnings moments by hours restriction and sample period

	Lower		Middle		Higher	
Sample	$\mathcal{P}$	$\mathcal{T}$	$\mathcal{P}$	$\mathcal{T}$	$\mathcal{P}$	$\mathcal{T}$
2013–2023, hours may change	8.93	2.05	8.64	3.06	8.39	4.06
2013–2023, unchanged hours	4.03	2.24	4.68	3.23	5.13	4.22
2017–2023, unchanged hours	6.08	2.83	6.31	4.72	7.09	6.08

Notes: Survey-weighted monthly log moments scaled by 10^4; January changes excluded and squared changes trimmed at pooled 1st/99th percentiles.

I estimate these products at lags one to six within the same job and annual reference period, allowing hours to change and requiring both endpoint changes to pass the main trimming rule. In every income group, the joint test in Table C.2 rejects zero dependence at lags two through six. Earnings dynamics, reporting error or sample selection could each generate this additional dependence; the test establishes a departure from the benchmark without separating these sources.

Table C.2: Products of earnings changes at different lags

Lag in months	Lower	Middle	Higher
1	−2.046 (0.112)	−3.058 (0.130)	−4.060 (0.133)
2	0.546 (0.071)	0.375 (0.086)	−0.091 (0.090)
3	−0.343 (0.115)	0.104 (0.131)	0.942 (0.151)
4	0.217 (0.084)	0.048 (0.095)	−0.375 (0.091)
5	−0.171 (0.074)	−0.610 (0.099)	−0.801 (0.091)
6	−0.451 (0.107)	0.183 (0.144)	0.796 (0.151)
Joint p, lags 2–6 equal zero	< 0.001	< 0.001	< 0.001

Notes: Survey-weighted products scaled by 10^4, using available pairs at each lag. Household-clustered standard errors in parentheses condition on fitted residuals and trimming. Joint tests impose five restrictions.

D Persistence and Statistical Inference

D.1 The Persistence Specification

For each income group g, I estimate the monthly persistence of perceived risk in the SCE and of the earnings volatility proxy in the SIPP:

$$q_{it} = a_i + \phi_g q_{i,t-1} + \varepsilon_{it}, \tag{D.1}$$

where q_{it} denotes adjusted perceived variance or the SIPP proxy, and ϕ_g measures its persistence. The unit effect a_i is specific to an uninterrupted response spell in the SCE or a job spell in the SIPP. Hours may change within a SIPP job spell, but lags never connect observations from different jobs.

To construct the SIPP proxy, I regress adjusted monthly earnings changes on a constant and their one-month lag, pooling income groups and applying monthly survey weights to retained pairs of consecutive changes. The coefficient on the lagged change is −0.2588. Squaring the residuals gives q_{it}, which measures the size of earnings movements unexplained by the previous month's change, regardless of sign. Without further trimming, this leaves 375,464 proxy observations in the lower-

income group, 623,762 in the middle-income group and 779,812 in the higher-income group. I preserve gaps in the monthly records when constructing subsequent lags.

I estimate equation (D.1) by system GMM, combining the equation in first differences with the equation in levels. Differencing removes the unit effect, but the lagged regressor $\Delta q_{i,t-1}$ and the differenced error $\Delta\varepsilon_{it} = \varepsilon_{it} - \varepsilon_{i,t-1}$ both contain the previous month's shock, $\varepsilon_{i,t-1}$. I therefore instrument the lagged regressor with earlier levels of q_{it}. The second lag is valid if the level errors are serially uncorrelated; serial correlation can require instruments further in the past.

The main specification uses $q_{i,t-3}$ and $q_{i,t-4}$ as instruments in the differenced equation. These lags leave at least one additional month of separation relative to the second lag, although their validity still depends on the error process. In the level equation, I instrument $q_{i,t-1}$ with $\Delta q_{i,t-1}$. The identifying moment restrictions are

$$\mathbb{E}[q_{i,t-j}\,\Delta\varepsilon_{it}] = 0, \qquad j \in \{3,4\}, \tag{D.2}$$

$$\mathbb{E}[\Delta q_{i,t-1}(a_i + \varepsilon_{it})] = 0. \tag{D.3}$$

The level restriction additionally requires changes in q_{it} to be uncorrelated with the unit effect. I collapse the instruments into one column per lag and use two-step estimation. All lags refer to calendar months within the relevant response or job spell. For GMM estimation, I hold each unit's survey weight fixed at its first available value.

D.2 Alternative Instruments and Numeracy

The opposing income patterns in persistence survive changes to the instrument window (Table D.1). Moving the window one month earlier or later leaves perceived risk more persistent at lower incomes and the SIPP proxy more persistent at higher incomes. The SCE pattern also holds for raw perceived variance and among high-numeracy respondents, whose perceived risk remains more persistent at lower incomes even though their adjusted mean-risk gap is close to zero.

Table D.1: Sensitivity of the persistence estimates

Specification	Lower	Middle	Higher
A. SCE perceived variance			
Adjusted, instruments at lags 2–3	0.346 (0.044)	0.108 (0.032)	0.139 (0.022)
Adjusted, lags 3–4 (main)	0.343 (0.046)	0.142 (0.026)	0.144 (0.025)
Adjusted, lags 4–5	0.350 (0.044)	0.105 (0.038)	0.112 (0.034)
Raw variance, lags 3–4	0.355 (0.042)	0.128 (0.028)	0.124 (0.034)
Adjusted, high numeracy, lags 3–4	0.274 (0.075)	0.102 (0.043)	0.126 (0.037)
B. SIPP volatility proxy			
Instruments at lags 2–3	0.193 (0.012)	0.206 (0.010)	0.275 (0.009)
Lags 3–4 (main)	0.193 (0.013)	0.187 (0.010)	0.235 (0.009)
Lags 4–5	0.237 (0.015)	0.239 (0.012)	0.279 (0.010)

Notes: Two-step system-GMM monthly AR(1) estimates. Standard errors in parentheses cluster by response spell and month (SCE) or job spell (SIPP). Only differenced-equation instruments vary across lag windows.

D.3 Standard Errors and Income-Group Comparisons

For SCE levels, I estimate group means and their covariance by regressing perceived variance on income-group indicators with survey weights and respondent-clustered standard errors. For the SIPP moments in Table 2, I calculate analytical standard errors clustered by household, retaining the covariance between the estimated second moment and adjacent-change product when constructing permanent, transitory and twelve-month variances. These calculations hold fitted earnings residuals, group assignments, sample membership and trimming thresholds fixed.

For SCE persistence, I cluster by response spell and calendar month and subtract their intersection, retaining the covariance induced by shared months when comparing income groups. For SIPP persistence, I cluster by job spell, with no overlapping clusters across income groups. The persistence standard errors condition on the fitted GMM weighting matrix and, for SIPP, the preliminary earnings regression. I use the joint covariance matrix to test equality of the higher- and lower-income estimates in Tables 2 and 3 with a one-restriction Wald test. Inference treats the preliminary adjustments as fixed and uses the stated clustering schemes without Census replicate weights.

E Accounting for Variation across Pay Components

E.1 Additive Contributions to Log Earnings Changes

I construct each pay component's contribution so that the contributions sum to the adjusted change in log total earnings. Let W_{it} denote total recorded pay and W_{it}^k component k. I first allocate adjusted pay using the recorded shares, $s_{it}^k = W_{it}^k / W_{it}$:

$$\widetilde{W}_{it} = \exp(z_{it}), \qquad \widetilde{W}_{it}^k = s_{it}^k \widetilde{W}_{it}, \qquad \sum_k \widetilde{W}_{it}^k = \widetilde{W}_{it}. \tag{E.1}$$

To express changes in these amounts as contributions to log earnings growth, I use the logarithmic mean of the positive totals:

$$M(a,b) = \frac{a-b}{\log a - \log b}, \qquad M(a,a) = a. \tag{E.2}$$

Dividing the change in component k by this common denominator gives

$$d_{it}^k = \frac{\widetilde{W}_{it}^k - \widetilde{W}_{i,t-1}^k}{M(\widetilde{W}_{it}, \widetilde{W}_{i,t-1})}. \tag{E.3}$$

The contributions sum exactly to $z_{it} - z_{i,t-1}$. Since I take logarithms only of total pay, the construction also retains observations in which a component is zero or the measured remainder is negative.

I use the benchmark's monthly changes, adjacent pairs and survey weights, with a common sample across components for each moment. Centring the adjacent products changes the bonus shares only slightly, from 35.7, 53.0 and 67.5 percent to 35.5, 52.9 and 67.3 percent. Household-clustered standard errors account for covariance with the group total and across income groups, holding fitted residuals, the sample and trimming thresholds fixed. The higher–lower bonus-share difference is 31.77 percentage points (SE 3.18).

E.2 Contributions to Permanent-Innovation Variation

I obtain each component's contribution to permanent-innovation variation using the same decomposition as for total earnings. Writing $\mathcal{S}_{kg} = \mathbb{E}[d_{it}^k d_{it} \mid g]$, this gives

$$\mathcal{P}_{kg} = \mathcal{S}_{kg} - 2\mathcal{T}_{kg}, \qquad \sum_k \mathcal{P}_{kg} = \mathcal{P}_g. \tag{E.4}$$

Wages and salaries excluding itemised payments account for 81.8–86.2 percent of permanent-innovation variation (Figure E.1), while the bonus share rises from 4.7 to 12.4 percent across income groups. Each contribution includes comovements with other pay categories as well as variation in the component itself, so it cannot be interpreted as a separate shock variance.

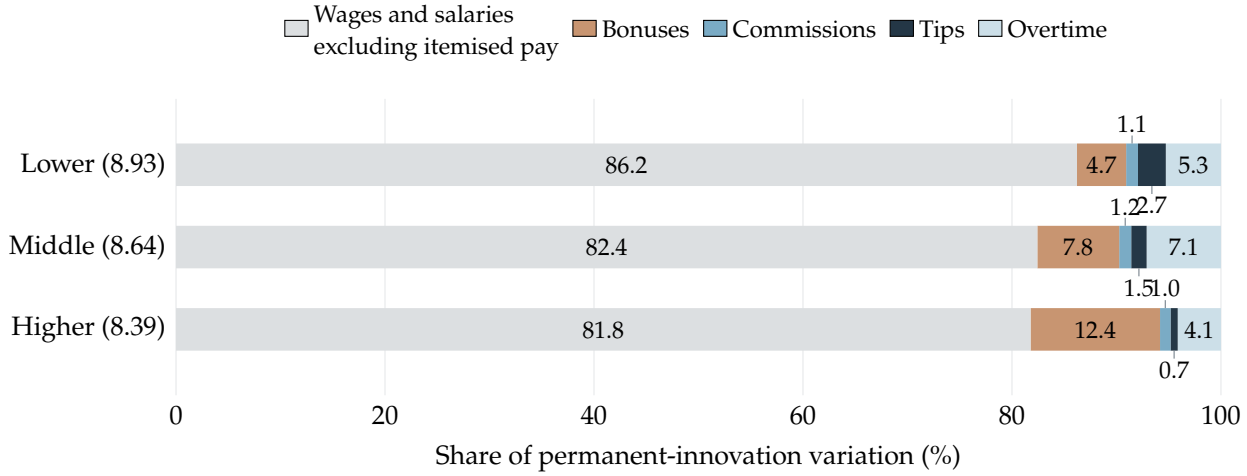


Figure E.1: Pay components' shares of permanent-innovation variation

Notes: SIPP, 2013–2023; hours-may-change sample in Table 2. Shares include cross-category terms. Parentheses give group totals in log moments scaled by 10^4.

F Payment Duration

F.1 Specification

To study how payments evolve after a break in receipt, I use months in which the worker received none of the relevant component in either of the preceding two months. I retain both positive and zero current payments and place no restriction on future receipt. The two-month gap excludes ongoing monthly payments while retaining payments that recur every quarter, so the profiles apply to months following a break in receipt.

In equation (5), I control for twelve lags of total hourly pay, the component's hourly amount and receipt indicator, and the combined hourly amount of the other itemised payments. I scale the amounts by the same average prior hourly earnings used for the dependent variable. The controls C_{it} also include log average prior hourly earnings and log hours in the preceding month, alongside calendar-date and income-group effects. I estimate the regressions with survey weights and standard errors clustered by worker. At $h = 0$, the current payment appears on both sides of the regression, giving $\beta_0^k = 1$ by construction.

I use a common sample across horizons, requiring valid, non-imputed observations in the same first recorded job from $t - 12$ through $t + 6$. Hours may change, but I exclude January initial months

and positive normalised initial payments above the within-income-group 99th percentile. Alongside zero-payment observations, the main sample contains 3,499 positive bonus payments from 2,192 workers and 514 positive overtime payments from 380 workers. The lower-income commission sample contains only seven positive payments from five workers, limiting the precision of separate income-group estimates for this component.

F.2 Alternative Payment Histories and Other Components

Bonuses fall away quickly and overtime payments persist when I require one, two or three preceding months without receipt (Table F.1). In each sample, the bonus coefficient falls close to zero immediately and the overtime coefficient remains positive through month six, although the estimated size of later overtime payments varies across samples.

Table F.1: Payment profiles under alternative receipt histories

Component	Prior zero months	Month 1	Month 3	Month 6
Bonuses	1	−0.011 (0.008)	0.035 (0.009)	0.045 (0.011)
	2	−0.010 (0.008)	0.034 (0.009)	0.044 (0.011)
	3	−0.010 (0.008)	0.015 (0.007)	0.032 (0.010)
Overtime	1	0.564 (0.076)	0.335 (0.088)	0.265 (0.102)
	2	0.579 (0.082)	0.329 (0.092)	0.251 (0.104)
	3	0.620 (0.086)	0.260 (0.078)	0.118 (0.036)
Commissions	2	−0.025 (0.016)	0.449 (0.089)	0.392 (0.105)

Notes: Equation (5); worker-clustered standard errors in parentheses. Upper-1% trimming thresholds vary by income group and receipt-history sample.

The size of the overtime coefficients is sensitive to large initial payments. Without trimming, the two-month-history coefficient falls to 0.21 at month one and 0.07 at month six, compared with 0.58 and 0.25 in the main specification. In the untrimmed regression, one worker accounts for nearly half the identifying variation. As a separate check, I hold the three-month sample's trimming thresholds fixed across receipt histories. Overtime payments continue to persist under this rule, with a month-six coefficient of 0.24 for the two-month history.

Commissions display peaks at months three and six, consistent with quarterly recurrence. The two-month-history sample contains 248 positive payments from 114 workers. Requiring three prior zero months excludes regular quarterly payments and weakens the month-three coefficient. For tips, only 11 positive payments from seven workers remain under the main rule, providing too little information for a reliable duration estimate.

F.3 Twelve-Month Payment Profiles

To examine whether payments recur over the year, I extend equation (5) through month twelve and re-estimate every horizon on a common sample within each component. I retain the same controls, weights and trimming rule, but now require valid, non-imputed records in the same job from $t - 12$ through $t + 12$. This longer observation window reduces the sample, so the estimates over the first six months can differ from those in Figure 3.

Figure F.1 shows a pronounced annual bonus peak, with a month-twelve coefficient of 0.433 (SE 0.041), based on 2,133 positive starts from 1,329 workers. The commission and overtime samples

contain fewer positive starts: 141 from 54 workers for commissions and 331 from 242 workers for overtime. The later overtime coefficients are imprecisely estimated.

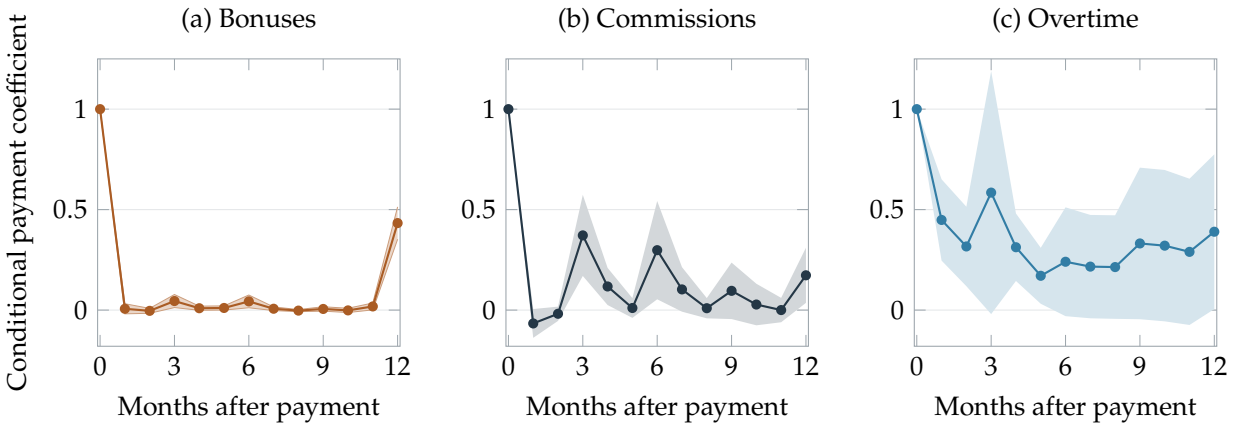


Figure F.1: Payment profiles through twelve months

Notes: SIPP, 2013–2023, pooled income groups. Coefficients relate future to current normalised hourly payments; $\beta_0 = 1$. Shading gives pointwise 95% confidence intervals, clustered by worker.

G Annual Recurrence in Pay Components

G.1 Specification and Inference

I apply equation (6) to each pay component within each income group to measure how much last year's payment predicts current receipt or amounts after accounting for recent payments. For component k, let Y_{it}^k denote either an indicator for a positive payment or the monthly amount in nominal dollars, and define the preceding eleven-month average as $\bar{Y}_{it}^k = \frac{1}{11}\sum_{s=1}^{11} Y_{i,t-s}^k$. The fixed effects absorb stable differences across worker–job matches and common calendar month–year effects. I include last month's payment separately from the eleven-month average to allow it to carry additional information. Conditional on these controls, γ_{kg} measures the association with the payment in the same month a year earlier. Multiplying the coefficient by 100 gives percentage points per prior receipt in the indicator regression, or dollars per \$100 more last year in the amount regression.

The sample requires positive total earnings and survey weights, observed amounts for all four itemised components, and a complete preceding twelve-month history on the same first recorded job. January outcomes are excluded, and hours may change. Amount regressions include zero-payment months and use the recorded nominal dollar amounts without additional trimming or rescaling. All four components use the same observations within each income group: 142,237 at lower incomes, 255,594 at middle incomes and 321,742 at higher incomes.

I estimate the regressions with survey weights and standard errors clustered by worker. To compare the higher- and lower-income annual coefficients, I use a joint worker-clustered covariance matrix that preserves dependence across equations; each reported p-value comes from a separate one-restriction Wald test without adjustment for multiple testing. The coefficients describe recurrence in recorded payments and may be affected by dynamic fixed effects, finite histories and survey reporting. Establishing what workers know or how accurately they forecast would require evidence beyond these descriptive regressions.

G.2 Comparison across Components

Only bonuses have positive annual coefficients after controlling for recent payment history (Table G.1). The negative coefficients for commissions, tips and overtime describe the additional association with last year's payment, conditional on recent receipts or amounts. These components can still be predictable at other frequencies, as illustrated by the quarterly recurrence of commissions in Section 3.2.

Table G.1: Annual recurrence across pay components

Component	Lower	Middle	Higher	Higher–lower p-value
A. Receipt probability (pp) per receipt last year				
Bonuses	6.90 (3.40)	7.52 (1.78)	20.59 (1.34)	< 0.001
Commissions	−5.40 (1.59)	−5.50 (1.35)	−7.95 (1.56)	0.253
Tips	−3.76 (1.73)	−2.28 (0.78)	−3.68 (1.20)	0.970
Overtime	−5.62 (1.04)	−6.94 (0.68)	−6.96 (0.61)	0.266
B. Amount ($) per $100 more last year				
Bonuses	2.75 (2.29)	7.34 (5.61)	29.83 (3.22)	< 0.001
Commissions	−9.35 (5.87)	−9.56 (3.51)	−19.43 (3.33)	0.136
Tips	−0.24 (0.16)	−2.30 (2.08)	−12.72 (4.83)	0.010
Overtime	−2.00 (0.90)	−3.98 (1.73)	−6.19 (1.19)	0.005

Notes: Annual coefficients from equation (6), multiplied by 100; worker-clustered standard errors in parentheses. p-values are higher–lower Wald tests.

G.3 Recurrence at Different Lags

To assess whether the higher-income advantage is specific to annual recurrence, I repeat the bonus regressions at three, six and twelve months on the annual specification's sample. Let Y_{it} denote either bonus receipt or its monthly nominal amount, including zeros. For each lag $h \in \{3, 6, 12\}$ and income group g, I estimate

$$Y_{it} = c^{(h)}_{i,j(i,t)} + \tau^{(h)}_t + \gamma_{gh} Y_{i,t-h} + \eta_{gh} Y_{i,t-1} + \zeta_{gh} \bar{Y}^{(h)}_{it} + e^{(h)}_{it}, \tag{G.1}$$

where $\bar{Y}^{(h)}_{it} = \frac{1}{h-1} \sum_{s=1}^{h-1} Y_{i,t-s}$ averages payments over the preceding two, five or eleven months. The fixed effects, survey weights and worker clustering follow the annual specification. Because the history average changes with h, I also estimate a receipt regression containing all three lags, last month's receipt and mean receipt over lags one to eleven excluding three and six. This holds the controls fixed across lag coefficients while preserving the sample, fixed effects, weights and clustering. Among these lags, the higher-income advantage appears only at twelve months in both receipt specifications and in the separate amount regressions (Table G.2).

G.4 Bonus Timing and Temporary Earnings Variation

To measure how much temporary earnings variation occurs in months when a bonus might recur, I apply the component accounting in Section 3.1 to months with bonus receipt in the same job one year earlier (Table 5). For each pair of adjacent adjusted earnings changes, I calculate

$$c^b_{it} = -\tfrac{1}{2} \left(d^b_{it} d_{i,t-1} + d_{it} d^b_{i,t-1} \right).$$

Table G.2: Bonus recurrence at three, six and twelve months

Lag	Lower	Middle	Higher	Higher–lower p-value
A. Separate lag regressions				
Receipt probability (pp) per receipt at the indicated lag				
3 months	21.18 (2.70)	19.46 (1.72)	14.37 (1.10)	0.020
6 months	9.79 (2.12)	10.82 (1.29)	8.01 (0.84)	0.436
12 months	6.90 (3.40)	7.52 (1.78)	20.59 (1.34)	< 0.001
Amount ($) per $100 more at the indicated lag				
3 months	2.41 (2.79)	−0.40 (0.91)	−3.51 (1.47)	0.061
6 months	−1.75 (2.24)	3.36 (5.43)	−6.17 (1.31)	0.089
12 months	2.75 (2.29)	7.34 (5.61)	29.83 (3.22)	< 0.001
B. Joint lag regression				
Receipt probability (pp) per receipt at the indicated lag				
3 months	20.62 (2.54)	18.35 (1.52)	13.90 (1.01)	0.014
6 months	9.14 (1.69)	9.91 (0.98)	5.99 (0.62)	0.080
12 months	5.97 (2.97)	6.75 (1.55)	20.13 (1.24)	< 0.001

Notes: Common sample across regressions; amounts include zeros. Worker-clustered standard errors in parentheses; p-values are higher–lower Wald tests.

I attribute this contribution to the middle month $t-1$ and record whether the worker received a bonus twelve months before that month. I retain the benchmark trimming and reporting-seam exclusions, allow hours to change and restrict the sample to observations with the required same-job history. I also require enough history to determine receipt within one month of the prior-year date, keeping the sample common to the exact-date and wider-window calculations.

The first row of Table 5 reports the survey-weighted fraction of eligible observations with prior-year receipt. For the remaining rows, I multiply c_{it}^b by the prior-receipt indicator and average over all eligible observations. I then divide by the survey-weighted mean of either c_{it}^b or $-d_{it}d_{i,t-1}$ in the same restricted sample. These ratios give the share of bonus variation or total temporary variation associated with prior-year receipt. They describe the concentration of measured variation in potentially recurring months, while leaving workers' anticipation of those payments unmeasured.

I calculate analytical standard errors by the delta method, allowing for joint uncertainty in each ratio's numerator and denominator and clustering by worker. Higher–lower Wald tests use the joint covariance of the group estimates. I hold survey weights, earnings adjustments, trimming and income classifications fixed in these calculations.

H Forecasting Earnings with Bonus Information

H.1 Specification and Evaluation

In equation (7), I forecast log total hourly earnings, including all pay components. The baseline uses current earnings, their three exact monthly lags and the arithmetic mean of all earlier log hourly earnings observed in the same recorded job, with separate month-of-year and year effects. I then add the current bonus share, its three exact monthly lags and its mean over all earlier observations in that job. The historical means exclude the current month and summarise older records; a separate twelve-month bonus lag is not included.

I require positive total pay and hourly earnings, valid survey weights, and observed bonus and commission amounts. I use commission records only to determine sample eligibility. Workers must have three consecutive prior monthly observations and remain observed in the same recorded job through the forecast endpoint. I retain months without bonuses and allow hours to change. At the twelve-month horizon, the comparison spans different annual reference periods.

I estimate each regression by survey-weighted least squares, separately by income group and horizon. I divide workers into five fixed sets, estimate the regression on four sets and predict earnings for workers in the remaining set, keeping all observations for a worker together. Both specifications use the same observations, weights and splits. Income-group classifications use the full observed panel, while coefficients and calendar effects are estimated using other workers throughout the sample period. The comparison consequently measures the predictive information in the records retrospectively.

I pool predictions for the excluded workers to calculate survey-weighted mean squared errors. To estimate uncertainty in their ratio, I use the delta method with worker clustering and retain the covariance between the two mean squared errors. The standard errors treat fitted forecasts as fixed, excluding uncertainty from re-estimating the regressions. Higher–lower Wald tests use the corresponding standard error of the difference in percentage gains. Because the forecasts use estimated coefficients, additional regressors can worsen predictions for excluded workers and produce negative gains. An optimal forecaster could avoid such losses by ignoring unhelpful information.

H.2 Current Information, Payment History and Additional Comparisons

Current bonus information accounts for most of the near-term forecast gain (Table H.1). At higher incomes, adding past bonus shares to the total-pay baseline reduces three-month mean squared error by 0.18 percent in panel A. Adding the current share then reduces the remaining error by 3.65 percent in panel B. These percentages refer to successive baselines and therefore cannot be added. At twelve months, the smaller overall gain comes mainly from past bonus records.

To check whether differences in the evaluation sample explain the pattern across horizons, panel C uses only observations for which all three future endpoints are available. I retain the original fitted forecasts in this common sample. The gains remain larger at higher incomes and decline with the forecast horizon.

H.3 Twelve-Month Forecasts with All Pay Components

Table H.2 extends the twelve-month forecast comparison to all four itemised pay components, adding each component's current share of total pay, three monthly lags and average share over all earlier observations in the same job. All specifications use the same total-pay baseline, observations, survey weights and worker splits. Adding all four pay components raises the higher-income forecast gain from 0.21 to 0.43 percent, an additional 0.22 percentage points ($p = 0.025$). The annual gains nevertheless remain small, and the higher–lower difference in the all-component specification is not statistically significant ($p = 0.122$).

Table H.1: Forecast gains under alternative information sets and evaluation samples

Forecast horizon	Lower	Middle	Higher
A. Past bonus shares added to total-pay information			
3 months	0.00 (0.01)	0.02 (0.01)	0.18 (0.02)
6 months	−0.01 (0.01)	0.04 (0.02)	0.14 (0.03)
12 months	−0.01 (0.01)	−0.05 (0.04)	0.19 (0.06)
B. Current bonus share added after past bonus shares			
3 months	0.40 (0.12)	0.94 (0.12)	3.65 (0.17)
6 months	0.15 (0.06)	0.34 (0.06)	1.67 (0.10)
12 months	0.00 (0.00)	0.02 (0.01)	0.03 (0.01)
C. Bonus information, common evaluation observations across horizons			
3 months	0.34 (0.08)	0.62 (0.14)	2.59 (0.18)
6 months	0.14 (0.04)	0.22 (0.06)	1.18 (0.11)
12 months	−0.01 (0.01)	−0.04 (0.04)	0.21 (0.06)

Notes: Percentage reductions in survey-weighted mean squared forecast error. Approximate worker-clustered standard errors in parentheses condition on fitted forecasts. Samples vary by horizon in A–B and are common in C.

Table H.2: Twelve-month forecast gains from additional pay components

Information added	Lower	Middle	Higher
A. Reduction in mean squared forecast error (%)			
Bonuses	−0.01 (0.01)	−0.04 (0.04)	0.21 (0.06)
All four components	−0.08 (0.31)	−0.05 (0.10)	0.43 (0.12)
B. Additional gain from all four components over bonuses alone			
Gain (pp)	−0.07 (0.31)	−0.01 (0.09)	0.22 (0.10)
Paired p-value	0.820	0.877	0.025

Notes: Both panels use total-pay baseline mean squared error as the denominator. Approximate worker-clustered standard errors in parentheses condition on fitted forecasts; panel B uses paired losses.

I Proofs of the Propositions

I.1 Proof of Proposition 1

To prove Proposition 1, I express earnings at the forecast date in terms of current permanent pay and future shocks:

$$w_{t+12} = p_t + \sum_{h=1}^{12} \psi_{t+h} + u_{t+12} + b_{t+12}. \tag{I.1}$$

Future innovations have mean zero conditional on the parameters and the full history before their arrival, so they are uncorrelated with p_t and earlier innovations even though their squares can be dependent. The future temporary components also have zero conditional means and zero covariance with the other terms. These covariances remain zero when I average over the worker's uncertainty about states and parameters, giving

$$V^j_{t,12} = P^j_t + \mathbb{E}[\sigma_u^2 + \sigma_b^2 \mid \mathcal{I}^j_t] + \sum_{h=1}^{12} \mathbb{E}[q_{t+h} \mid \mathcal{I}^j_t]. \tag{I.2}$$

For given q_{t+1} and ϑ, iterating the ARCH conditional mean gives

$$\mathbb{E}[q_{t+h} \mid q_{t+1}, \vartheta, \mathcal{I}_t^j] = \bar{q} + \alpha^{h-1}(q_{t+1} - \bar{q}). \tag{I.3}$$

Summing over $h = 1, \ldots, 12$ and averaging over the joint posterior gives equation (14), retaining dependence between α, $\bar{q}$ and q_{t+1}. Since future innovations have zero conditional mean at every parameter value, uncertainty about their variances enters the forecast through the expected conditional variance.

For recurring payments, substitute $b_{t+12} = v_{t+12} + D_{t+12}a_{t+12}$ in equation (I.1). Future innovations have mean zero conditional on the parameters, the payment schedule and the full history before their arrival, so the same covariance argument applies. The forecast decomposition becomes

$$\begin{aligned} V_{t,12}^j = P_t^j &+ \mathbb{E}[\sigma_u^2 + \sigma_v^2 + D_{t+12}\sigma_a^2 \mid \mathcal{I}_t^j] + 12\mathbb{E}[\bar{q} \mid \mathcal{I}_t^j] \\ &+ \mathbb{E}\Big[A_{12}(\alpha)(q_{t+1} - \bar{q}) \mid \mathcal{I}_t^j\Big]. \end{aligned} \tag{I.4}$$

The expectation $\mathbb{E}[D_{t+12}\sigma_a^2 \mid \mathcal{I}_t^j]$ averages jointly over the payment schedule and annual-payment variance. These can be dependent in the posterior, so the expectation need not equal $\Pr(D_{t+12} = 1 \mid \mathcal{I}_t^j)\mathbb{E}[\sigma_a^2 \mid \mathcal{I}_t^j]$. Under $\mathcal{I}_t^2$, the worker knows D_{t+12} and can take it outside the expectation. Setting $D_s = 0$ for every s and $\sigma_v^2 = \sigma_b^2$ recovers the baseline.

I.2 Proof of Proposition 2

Proposition 2 follows from the conditional law of total variance and the nesting $\mathcal{I}_t^0 \subseteq \mathcal{I}_t^1$:

$$\begin{aligned} \operatorname{Var}(w_{t+12} \mid \mathcal{I}_t^0) = \mathbb{E}[\operatorname{Var}(w_{t+12} \mid \mathcal{I}_t^1) \mid \mathcal{I}_t^0] \\ + \operatorname{Var}\big(\mathbb{E}[w_{t+12} \mid \mathcal{I}_t^1] \mid \mathcal{I}_t^0\big). \end{aligned} \tag{I.5}$$

The future shocks in equation (I.1) have zero conditional mean, so $\mathbb{E}[w_{t+12} \mid \mathcal{I}_t^1] = \widehat{p}_t^1$, and substitution gives the proposition. The reduction is strict when this forecast has positive conditional variance across the histories of b_t consistent with $\mathcal{I}_t^0$. The result averages over histories drawn under a common prior. Comparing income-group averages at different fixed true parameters requires a separate calculation.

The same argument establishes Proposition 2 with recurring payments, since the future innovations remain centred conditional on the parameters and payment schedule. In the comparison between $\mathcal{I}_t^1$ and $\mathcal{I}_t^2$, knowing the schedule similarly removes uncertainty on average when the possible schedules imply different forecasts of permanent pay. The worker may nevertheless report more risk on learning that an annual payment is due in the forecast month, because its amount remains uncertain. Under the same common earnings process and prior, the centred-innovation assumptions also give $\mathbb{E}[w_{t+12} \mid \mathcal{I}_t^2] = \widehat{p}_t^2$. Applying the conditional law of total variance to $\mathcal{I}_t^1 \subseteq \mathcal{I}_t^2$ therefore gives

$$\begin{aligned} V_{t,12}^1 - \mathbb{E}[V_{t,12}^2 \mid \mathcal{I}_t^1] &= \operatorname{Var}\big(\mathbb{E}[w_{t+12} \mid \mathcal{I}_t^2] \mid \mathcal{I}_t^1\big) \\ &= \operatorname{Var}(\widehat{p}_t^2 \mid \mathcal{I}_t^1) \geq 0. \end{aligned} \tag{I.6}$$

If knowing the schedule leaves the forecast of permanent pay unchanged, this average reduction is zero.

For the known-variance case with $\alpha = 0$ and no recurring payments, the posterior-variance recur-

sion is $P_t^j = f(P_{t-1}^j, R_j)$, where $f(P, R) = R(P + \bar{q})/(P + \bar{q} + R)$. Its derivatives are

$$\frac{\partial f}{\partial P} = \frac{R^2}{(P + \bar{q} + R)^2} \geq 0, \qquad \frac{\partial f}{\partial R} = \frac{(P + \bar{q})^2}{(P + \bar{q} + R)^2} \geq 0. \tag{I.7}$$

Since $R_1 \leq R_0$, a common initial permanent-pay variance implies $P_t^1 \leq P_t^0$ at every date by induction. All remaining terms in equation (14) are identical under the two information sets, yielding $V_{t,12}^0 - V_{t,12}^1 = P_t^0 - P_t^1 \geq 0$.

The conditional-variance argument also applies to beliefs about long-run permanent risk. Under the common prior and earnings process, and assuming $\mathbb{E}[\bar{q}^2] < \infty$, observing the temporary component separately reduces posterior uncertainty about $\bar{q}$ on average. Its posterior mean is a martingale by iterated expectations. Assessing the persistence of the worker's full earnings-risk report requires following the joint evolution of all its forecast components.

I.3 Proof of Proposition 3

To prove Proposition 3, return to the known-variance benchmark with $\alpha = 0$ and no recurring payments. Before each payment, the new permanent shock adds $\bar{q}$ to uncertainty about permanent pay. Substituting the Kalman gain into $P_t^j = (1 - K_t^j)(P_{t-1}^j + \bar{q})$ gives the recursion $P_t^j = f(P_{t-1}^j, R_j)$, where $f(P, R) = R(P + \bar{q})/(P + \bar{q} + R)$. For $P \geq 0$ and finite $R > 0$, the derivative in equation (I.7) satisfies $0 \leq \partial f/\partial P \leq [R/(\bar{q} + R)]^2 < 1$. The variance therefore converges from any finite initial value to a unique fixed point. At $R = 0$, the payment reveals permanent pay exactly and the posterior variance is zero.

At the fixed point, $P_\infty^j = R_j(P_\infty^j + \bar{q})/(P_\infty^j + \bar{q} + R_j)$, which rearranges to $(P_\infty^j)^2 + \bar{q}P_\infty^j = \bar{q}R_j$. The non-negative root gives P_∞^j in equation (16). The worker continues to revise her estimate of permanent pay as payments arrive, while the uncertainty around that estimate remains constant in steady state. With $\alpha = 0$ and known parameters, the remaining terms in Proposition 1 sum to $12\bar{q} + \sigma_u^2 + \sigma_b^2$, giving $V_{\infty,12}^j$.

Holding $\bar{q}$ and total temporary variance $R_0 > 0$ fixed, an increase in σ_b^2 reduces $R_1 = R_0 - \sigma_b^2$ by the same amount. The worker observing only total pay still faces noise variance R_0, while the worker observing b_t separately receives a more precise signal. The contribution of future shocks to forecast risk remains unchanged, so for $0 < \sigma_b^2 < R_0$,

$$\begin{aligned} \left.\frac{\partial V_{\infty,12}^1}{\partial \sigma_b^2}\right|_{\bar{q},R_0} &= -\frac{\bar{q}}{\sqrt{\bar{q}^2 + 4\bar{q}(R_0 - \sigma_b^2)}} < 0, \\ \left.\frac{\partial V_{\infty,12}^0}{\partial \sigma_b^2}\right|_{\bar{q},R_0} &= 0. \end{aligned} \tag{I.8}$$

The expressions are continuous at the endpoints. A larger separately observed share therefore lowers perceived risk under $\mathcal{I}_t^1$ and increases the reduction relative to total-pay information. This comparison holds total temporary variance fixed. If σ_b^2 rises while σ_u^2 is held fixed, uncertainty about future temporary pay also increases.

With known parameters and a known calendar, the filtering variance can converge to a repeating annual pattern when recurring payments remain in the pay signal. Subtracting the observed b_t leaves

constant noise variance σ_u^2. An unknown calendar is included in the joint posterior described in Appendix J.1.

I.4 Proof of Proposition 4

To prove Proposition 4, I construct the Gaussian approximation without recurring payments and use it to derive the gain and the persistence of estimation errors. I first write the ARCH transition as

$$q_t = (1-\alpha)\bar{q} + \alpha q_{t-1} + \alpha q_{t-1}(\varepsilon_{t-1}^2 - 1). \tag{I.9}$$

Conditional on q_{t-1} and the parameters, the final term has mean zero and variance $2\alpha^2 q_{t-1}^2$. The Gaussian approximation preserves this conditional mean and replaces the final term by $\eta_t \sim N(0, Q)$. The resulting approximation to permanent-pay risk is

$$h_t = (1-\alpha)\bar{q} + \alpha h_{t-1} + \eta_t.$$

Setting $Q = 2\alpha^2 \bar{q}^2$ also matches its variance at the long-run risk $\bar{q}$. The result holds for any common $Q > 0$, with $0 < \alpha < 1$.

The pay filter is evaluated at constant reference risk $\bar{q}$, with steady-state variance P_∞^j from Proposition 3 and temporary noise variance R_j. Write $\delta_{t-1}^j = p_{t-1} - \widehat{p}_{t-1}^j$ for its inherited estimation error and let r_t^j be the temporary noise remaining in y_t^j. The pay surprise is $e_t^j = y_t^j - \widehat{p}_{t-1}^j = \psi_t + \delta_{t-1}^j + r_t^j$. At the reference filter, approximate δ_{t-1}^j by a mean-zero Gaussian variable with variance P_∞^j, independent of current risk and new shocks. With $\psi_t \mid h_t \sim N(0, h_t)$ and $r_t^j \sim N(0, R_j)$, this gives $e_t^j \mid h_t \approx N(0, h_t + P_\infty^j + R_j)$ locally around positive risk $\bar{q}$.

For a mean-zero Gaussian variable X with variance v, $\mathbb{E}[X^2] = v$ and $\mathrm{Var}(X^2) = 2v^2$. Subtracting the reference contributions of inherited estimation error and temporary pay gives the risk reading $\widetilde{h}_t^j = (e_t^j)^2 - P_\infty^j - R_j$, whose conditional moments are

$$\begin{aligned} \mathbb{E}[\widetilde{h}_t^j \mid h_t] &\approx h_t, \\ \mathrm{Var}(\widetilde{h}_t^j \mid h_t) &\approx 2(h_t + P_\infty^j + R_j)^2. \end{aligned} \tag{I.10}$$

Evaluating the second expression at $h_t = \bar{q}$ gives the approximate noise variance

$$\Omega_j = 2(\bar{q} + P_\infty^j + R_j)^2. \tag{I.11}$$

Observing b_t separately lowers both P_∞^j and R_j, making the reading more precise. To obtain a Gaussian filter for risk, I approximate the reading by $s_t^j = h_t + \nu_t^j$, where $\nu_t^j \sim N(0, \Omega_j)$. This matches its first two conditional moments at the reference risk, although a squared Gaussian surprise is itself non-Gaussian. I also assume that state innovations and signal errors are independent across months and of each other. This simplifies the dependence present in the ARCH model, where the squared permanent shock affects both the pay surprise and next month's risk.

The Gaussian process for h_t approximates risk locally around a positive reference value and can take negative values outside that neighbourhood. I use it for the analytical illustration; the full model retains the ARCH process and its joint posterior. Since the risk reading is constructed from the payment already used by the pay filter, the joint calculation uses that payment once.

Let $S_t^j = \mathrm{Var}(h_t \mid s_1^j, \ldots, s_t^j)$ in the auxiliary Gaussian model. The worker knows $\bar{q}$, α, Q and Ω_j, and

the initial state h_0 has a Gaussian prior independent of all innovations and signal errors. The usual Kalman recursion gives

$$S_t^{j,-} = \alpha^2 S_{t-1}^j + Q, \qquad K_{h,t}^j = \frac{S_t^{j,-}}{S_t^{j,-} + \Omega_j}, \qquad S_t^j = (1 - K_{h,t}^j) S_t^{j,-}. \tag{I.12}$$

At the steady-state filtering variance, $S_\infty^j = K_h^j \Omega_j$. Substituting this into $S_\infty^j = (1 - K_h^j)(\alpha^2 S_\infty^j + Q)$ yields

$$\alpha^2 \Omega_j (K_h^j)^2 + \left[Q + (1 - \alpha^2)\Omega_j\right] K_h^j - Q = 0. \tag{I.13}$$

The left-hand side is strictly increasing in $K_h^j \geq 0$, negative at zero and positive at one. Its unique root in $(0, 1)$ is

$$K_h^j = \frac{2Q}{Q + (1 - \alpha^2)\Omega_j + \sqrt{\left[Q + (1 - \alpha^2)\Omega_j\right]^2 + 4\alpha^2 \Omega_j Q}}. \tag{I.14}$$

This gives the gain in equation (17). A larger gain increases the response to a given signal surprise. Differentiating equation (I.13) with respect to the signal's noise variance gives

$$\frac{\partial K_h^j}{\partial \Omega_j} = -\frac{\alpha^2 (K_h^j)^2 + (1 - \alpha^2) K_h^j}{2\alpha^2 \Omega_j K_h^j + Q + (1 - \alpha^2)\Omega_j} < 0. \tag{I.15}$$

For the composition comparison, hold $\bar{q}$, α, $R_0 > 0$ and $Q > 0$ fixed, with $R_1 = R_0 - \sigma_b^2$. Proposition 3 gives $\partial P_\infty^1 / \partial \sigma_b^2 < 0$. Thus

$$\begin{aligned} \frac{\partial \Omega_1}{\partial \sigma_b^2} &= 4(\bar{q} + P_\infty^1 + R_1)\left(\frac{\partial P_\infty^1}{\partial \sigma_b^2} - 1\right) < 0, \\ \frac{\partial K_h^1}{\partial \sigma_b^2} &= \frac{\partial K_h^1}{\partial \Omega_1}\frac{\partial \Omega_1}{\partial \sigma_b^2} > 0. \end{aligned} \tag{I.16}$$

For a positive variance σ_b^2, observing the component separately also implies $R_1 < R_0$ and $P_\infty^1 < P_\infty^0$, giving $\Omega_1 < \Omega_0$ and $K_h^1 > K_h^0$.

Finally, let $\xi_t^j = \widehat{h}_t^j - h_t$ denote the estimation error. Subtracting the state transition from the belief update gives

$$\xi_t^j = \alpha(1 - K_h^j)\xi_{t-1}^j - (1 - K_h^j)\eta_t + K_h^j \nu_t^j. \tag{I.17}$$

An earlier error enters the next estimate with coefficient $\alpha(1 - K_h^j)$. Holding subsequent state innovations and signal errors fixed, its contribution after k months is multiplied by $[\alpha(1 - K_h^j)]^k$, so a larger gain makes the error disappear more quickly. The estimated risk level has a different conditional-mean relation: $\mathbb{E}[\widehat{h}_t^j - \bar{q} \mid s_1^j, \ldots, s_{t-1}^j] = \alpha(\widehat{h}_{t-1}^j - \bar{q})$ under either information set in the Gaussian model. Faster correction of a past error can therefore coexist with the same conditional-mean persistence of estimated risk.

In the exact ARCH model with known parameters, iterated expectations give

$$\begin{aligned} \mathbb{E}[\widehat{q}_{t+2|t+1}^j - \bar{q} \mid \mathcal{I}_t^j] &= \mathbb{E}[q_{t+2} - \bar{q} \mid \mathcal{I}_t^j] \\ &= \alpha\left(\widehat{q}_{t+1|t}^j - \bar{q}\right). \end{aligned} \tag{I.18}$$

The second equality uses $\mathbb{E}[\psi_{t+1}^2 \mid q_{t+1}, \vartheta, \mathcal{I}_t^j] = q_{t+1}$. Under either information set, the worker's

expected permanent-shock variance follows this conditional-mean relation. The persistence of her full earnings-risk report also depends on how its other components evolve.

J Quantitative Implementation and Sensitivity

J.1 Joint Bayesian Updating

I obtain the joint update by first using the new payment to reweight the worker's beliefs about past permanent pay, current risk and the parameters, then inferring the current permanent shock. In the baseline without a known payment calendar, let Π^j_{t-1} denote her prior over $(p_{t-1}, q_t, \vartheta)$ given $\mathcal{I}^j_{t-1}$, and let $\varphi(x; m, v)$ be the Gaussian density. The payment likelihoods and posterior weights are

$$\begin{aligned} L^0_t &= \varphi(w_t; p_{t-1}, q_t + \sigma_u^2 + \sigma_b^2), \\ L^1_t &= \varphi(w_t - b_t; p_{t-1}, q_t + \sigma_u^2)\varphi(b_t; 0, \sigma_b^2), \\ d\widetilde{\Pi}^j_t &= \frac{L^j_t \, d\Pi^j_{t-1}}{\int L^j_t \, d\Pi^j_{t-1}}. \end{aligned} \tag{J.1}$$

For each candidate $(p_{t-1}, q_t, \vartheta)$, define $k^j_t = q_t/(q_t + R_j)$, where $R_0 = \sigma_u^2 + \sigma_b^2$ and $R_1 = \sigma_u^2$. Gaussian conditioning gives

$$\psi_t \mid p_{t-1}, q_t, \vartheta, \mathcal{I}^j_t \sim N\Big(k^j_t(y^j_t - p_{t-1}), (1 - k^j_t)q_t\Big). \tag{J.2}$$

I combine this conditional distribution with the transitions $p_t = p_{t-1} + \psi_t$ and $q_{t+1} = (1-\alpha)\bar{q} + \alpha\psi_t^2$ and integrate over $\widetilde{\Pi}^j_t$ to obtain the joint posterior Π^j_t over $(p_t, q_{t+1}, \vartheta)$. Averaging forecast moments over this posterior retains the dependence between states and parameters. The numerical implementation below approximates this update and treats annual payments separately when the worker knows the calendar.

J.2 Numerical Learning

Workers begin with prior distributions common across income groups. Under $\mathcal{I}^0$ and $\mathcal{I}^1$, let $\sigma_T^2 = \sigma_u^2 + \sigma_b^2$. The variance priors are

$$\begin{aligned} \log \bar{q} &\sim N\Big(\log \bar{q}_0 - \tfrac{1}{2}\tau_q^2, \tau_q^2\Big), \\ \log \sigma_T^2 &\sim N\Big(\log \sigma^2_{T,0} - \tfrac{1}{2}\tau_T^2, \tau_T^2\Big). \end{aligned} \tag{J.3}$$

I set $\bar{q}_0 = 8.65$ and $\sigma^2_{T,0} = 3.06$ from the SIPP cross-group averages, and maintain $\tau_q = \tau_T = 2.5$. An independent share $s \sim U(0,1)$ assigns $\sigma_b^2 = s\sigma_T^2$ and $\sigma_u^2 = (1-s)\sigma_T^2$. Under $\mathcal{I}^2$, the same prior family and means apply to $\sigma_u^2 + \sigma_v^2$ and its irregular-bonus share. A separate lognormal prior for annual-payment variance has mean 40.7 and log standard deviation 1.5. Workers also learn α, starting from $U(0, 0.55)$. I treat these priors as assumptions and do not fit their parameters to the SCE.

I approximate the worker's joint beliefs with a resample–move particle filter, using 1,024 particles initialised at scrambled Sobol points of the common prior. Each particle represents a possible permanent-pay history, current risk and parameter vector. After a payment arrives, the Gaussian signal update proposes current permanent pay conditional on each particle, and the predictive likelihood determines its weight. Under $\mathcal{I}^1$, this likelihood also uses the separately observed bonus. Under $\mathcal{I}^2$,

I estimate irregular-bonus variance using only months without an annual payment, while subtracting the observed bonus from total pay in every month when filtering permanent pay.

Before systematic resampling, I calculate perceived earnings risk from the weighted conditional moments. Permanent shocks contribute

$$\sum_{h=1}^{12} \mathbb{E}[q_{t+h} \mid \mathcal{I}_t^j] = \mathbb{E}\Big[12\bar{q} + A_{12}(\alpha)(q_{t+1} - \bar{q}) \mid \mathcal{I}_t^j\Big]. \tag{J.4}$$

I take this expectation over the joint posterior of states and parameters, preserving the dependence between α and current risk. After resampling, I use four Metropolis–Hastings moves to update the parameters and permanent-pay path. Forward filtering and backward sampling under a constant-variance reference process provide the path proposals, and the acceptance probability corrects for ARCH risk. Under $\mathcal{I}^0$, each move changes total temporary variance and redraws its unidentified split from the conditional prior.

I give initial permanent pay the prior $N(0, 10{,}000)$. For the true earnings histories, I initialise risk using 256 ARCH burn-in draws. To represent the worker's uncertainty about initial risk, the filter approximates its stationary distribution using 48 auxiliary draws per particle, which I also update in the Metropolis–Hastings step.

I update annual-payment variance separately from the particle filter. Let $z = \log \sigma_a^2$ and write $\widehat{\sigma}_{v,t}^2 = \mathbb{E}[\sigma_v^2 \mid \mathcal{I}_t^2]$. In each scheduled payment month, I evaluate its density on 4,000 equally spaced points between $\log(10^{-3})$ and $\log(10^5)$:

$$\pi_t(z) \propto \exp\left[-\frac{(z-\mu_a)^2}{2\tau_a^2}\right] \prod_{\substack{s \le t \\ D_s = 1}} \varphi\Big(b_s; 0, \widehat{\sigma}_{v,t}^2 + e^z\Big), \qquad \mu_a = \log(40.7) - \frac{1.5^2}{2}, \quad \tau_a = 1.5, \tag{J.5}$$

where $\varphi(\cdot; m, v)$ denotes the Gaussian density. I use the unrounded cross-group mean for the prior centre and obtain the posterior mean of e^z from the normalised grid weights. Annual-payment variance contributes to reported risk only when $D_{t+12} = 1$, which coincides with $D_t = 1$ under the fixed annual schedule. The calculation approximates joint inference about the two variances by holding irregular variance at its current mean and estimating it from months without annual payments.

Without a calendar, the worker includes estimated bonus variance in every forecast. With a calendar, she includes annual variance only when the forecast month contains the scheduled payment, as in equation (13). This timing effect arises because she forecasts earnings in month $t + 12$; a rolling twelve-month earnings total would always contain an annual payment. I assume that workers under $\mathcal{I}^0$ and $\mathcal{I}^1$ do not learn the calendar. The numerical comparison consequently changes both their information and their forecasting model, while the information result in Appendix I.2 holds the process and prior common.

J.3 Calibration Checks

I calibrate the earnings process using the same-job SIPP sample with changing hours, setting $\bar{q}_g = \mathcal{P}_g$, $\sigma_{b,g}^2$ equal to the bonus contribution and $\sigma_{u,g}^2 = \mathcal{T}_g - \sigma_{b,g}^2$. A share ρ of workers receives an annual payment in a fixed month, which I distribute uniformly across recipients. Averaging over workers and calendar months gives

$$\sigma_v^2 + \frac{\rho}{12}\sigma_a^2 = \sigma_b^2. \tag{J.6}$$

This identity allocates the measured bonus variation between irregular and annual payments.

Bonus timing is measured among workers with enough same-job history to observe the corresponding month one year earlier. Let π_g be the share of months with bonus receipt one year earlier, and let B_g^{yes} and B_g^{no} denote the bonus variance contributions in those months and the remaining months. Writing B_g^{lb} for the average contribution in this lookback sample, the mapping is

$$\rho_g = 12\pi_g, \qquad \sigma^2_{a,g} = \kappa_g(B_g^{\text{yes}} - B_g^{\text{no}}), \qquad \sigma^2_{v,g} = \kappa_g B_g^{\text{no}}, \qquad \kappa_g = \frac{\sigma^2_{b,g}}{B_g^{\text{lb}}}. \tag{J.7}$$

Since a share ρ_g of workers receives one payment each year, the monthly frequency of annual payments is $\rho_g/12$. I use the excess variation in recurring months to set annual-payment variance and scale both variance components by κ_g to reconcile the sample with prior-year histories with the full calibration sample. This ensures that equation (J.6) holds. The fixed annual schedule approximates the observed recurrence; workers' knowledge of the schedule is a separate modelling assumption.

To calibrate α, I match persistence in the earnings volatility proxy, which reflects both permanent risk and the arrival and disappearance of temporary payments. I simulate 2,048 workers per income group for 47 months at each candidate $\alpha \in \{0.30, 0.32, \ldots, 0.50\}$. I then pool earnings changes across groups, remove their linear dependence on the previous change and apply the main system GMM specification to twelve-month windows of squared residuals. I weight each squared difference between simulated and SIPP persistence by the inverse squared SIPP standard error. Table J.1 shows that the objective is relatively flat between 0.42 and 0.46. The selected value approximates overall persistence, although the simulated coefficients remain similar across incomes and leave the empirical income gradient unexplained.

Table J.1: Calibration to earnings-risk persistence

	Lower	Middle	Higher	Weighted distance
SIPP	0.193	0.187	0.235	–
Simulated proxy by α				
0.30	0.138	0.161	0.169	77.8
0.42	0.195	0.200	0.200	16.4
0.44	0.206	0.207	0.206	15.2
0.46	0.217	0.215	0.212	17.6

Notes: Three-group earnings process, independent of workers' beliefs. SIPP standard errors: 0.013, 0.010 and 0.009. The income-dependent calibration retains $\alpha = 0.44$.

I use the same proxy construction and instruments in the simulation and the data, with differences in how observations enter estimation. Simulated windows are balanced, equally weighted and untrimmed; SIPP observations have survey weights, trimming and job spells of varying length. Gaussian innovations with ARCH risk also allow large earnings movements in the simulation.

J.4 Continuous Income and Smooth Parameter Schedules

To let pay composition vary smoothly as workers move across the income distribution, I use anchor incomes of \$35,100, \$73,788 and \$158,796. I choose parameter values at these anchors so that their averages within the three job-start income groups reproduce the SIPP targets. Between the anchors, I

interpolate the logarithms of $\bar{q}$, σ_b^2, σ_u^2, ρ and σ_a^2 using a shape-preserving cubic in log income. Beyond the anchors, I assume that each log schedule follows a bounded hyperbolic-tangent tail preserving its endpoint slope, with a scale of 0.5 log-income units.

Starting log annual income follows a normal distribution with mean 11.48849 and standard deviation 0.754. I draw 8,192 workers within each income stratum and weight them by the implied population shares.

In month t, I evaluate other temporary variance, irregular-bonus variance and annual-payment variance at the worker's permanent income entering the month. Irregular-bonus variance follows $\sigma_v^2(x) = \sigma_b^2(x) - \rho(x)\sigma_a^2(x)/12$. Long-run permanent variance, annual-payment eligibility and the payment month remain fixed at job entry, and $\alpha = 0.44$ is common. A worker's annual-payment eligibility therefore stays the same as her income changes, so the initial bonus-variance identity need not hold at her subsequent income. Her temporary-pay variances evolve smoothly even when crossing an income threshold changes her classification.

I approximate workers' learning about the evolving process by allowing them to update estimates of constant temporary variances from their payments. They do not use income or learn the variance schedules. Their forecasts combine posterior mean temporary variances, the annual update in equation (J.5) and the permanent-risk forecast in equation (J.4).

I check the effect of allowing temporary variances to change with income using identical simulated histories. Relative to fixed composition, the group earnings-variance moments change by at most 2.3 percent, permanent-variance estimates change by at most 0.01, and the volatility-proxy persistence estimates remain similar. I therefore retain the calibrated schedules and $\alpha = 0.44$.

For the main results, I assign income groups using earnings over the twelve months ending at the first report, using own earnings to approximate household income at SCE entry. I hold groups fixed within each reporting window, but reassign them when the number of payments observed increases. For the temporary-payment experiment, I use pre-intervention annual earnings to fix the groups. For the policy experiment, I update them each month using the latest twelve payments excluding transfers. I retain the same income thresholds throughout, and each policy comparison follows the same workers with and without transfers.

J.5 Payments Observed and Persistence

I estimate persistence using twelve reports after payments $A, \ldots, A + 11$ from the same simulated workers, with the population weights and income groups described above. For each information set, I exclude workers with any report outside the pooled 1st/99th percentiles. Workers learn from the first payment, so A records the number of payments observed when I begin measuring their reports; it is not a measure of employment tenure. The benchmark uses $A = 48$, while the figures extend to $A = 72$ and require reports through payment 83. Table J.2 reports the benchmark levels and persistence under all three information sets.

Table J.2: Perceived risk by payment information

A. Mean risk relative to higher income

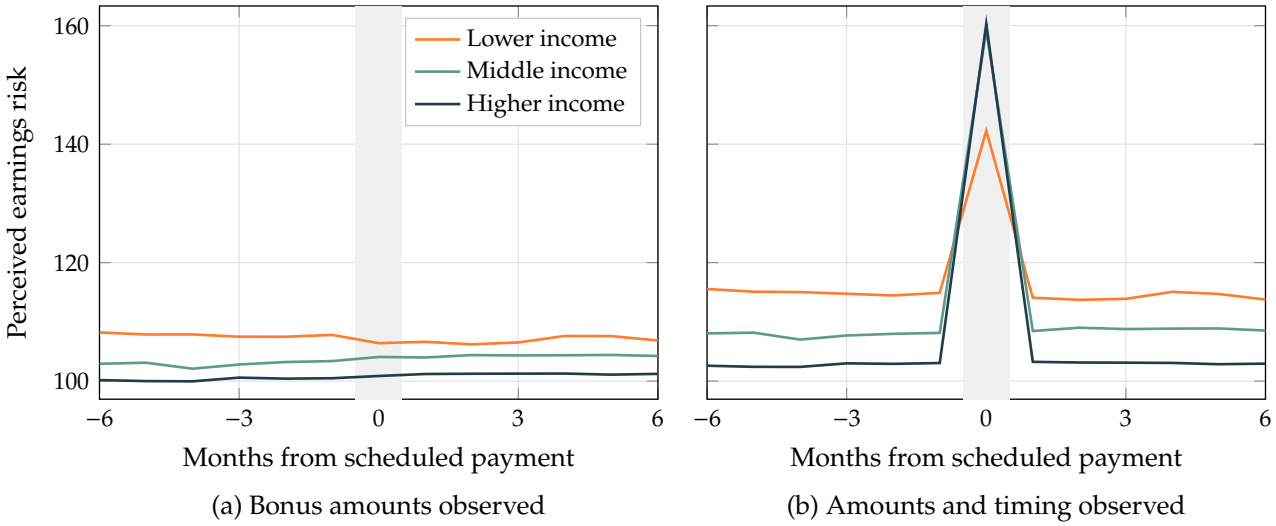
	Lower/higher	Middle/higher
SCE	1.81 (0.130)	0.99 (0.064)
Total pay, $\mathcal{I}^0$	0.95 (0.005)	0.97 (0.005)
Bonus amounts, $\mathcal{I}^1$	1.11 (0.005)	1.05 (0.005)
Amounts and timing, $\mathcal{I}^2$	1.11 (0.005)	1.05 (0.005)

B. Monthly persistence

	Lower	Middle	Higher	Lower−higher
SCE	0.343 (0.046)	0.142 (0.026)	0.144 (0.025)	0.199
Total pay, $\mathcal{I}^0$	0.401 (0.016)	0.392 (0.016)	0.445 (0.014)	-0.044 (0.021)
Bonus amounts, $\mathcal{I}^1$	0.310 (0.015)	0.351 (0.016)	0.346 (0.014)	-0.035 (0.020)
Amounts and timing, $\mathcal{I}^2$	0.282 (0.014)	0.244 (0.014)	0.179 (0.012)	0.104 (0.020)

Notes: Model estimates use population weights; groups use earnings in the twelve months ending at the first report. Workers with any report outside pooled 1st/99th percentiles are excluded by information set. Model standard errors in parentheses use 100 worker-bootstrap draws, conditional on calibration and assumptions. SCE inference follows Appendix D.3.

To show how the payment calendar affects perceived risk, I align annual-payment recipients around their scheduled month in Figure J.1. When workers know the timing, their risk rises briefly as the forecast includes an annual payment and falls back the following month. The annual-payment contribution in that month averages 28, 51 and 57 variance units in the lower-, middle- and higher-income groups. The population effect depends both on these changes among recipients and on the share of workers receiving annual payments.



(a) Bonus amounts observed

(b) Amounts and timing observed

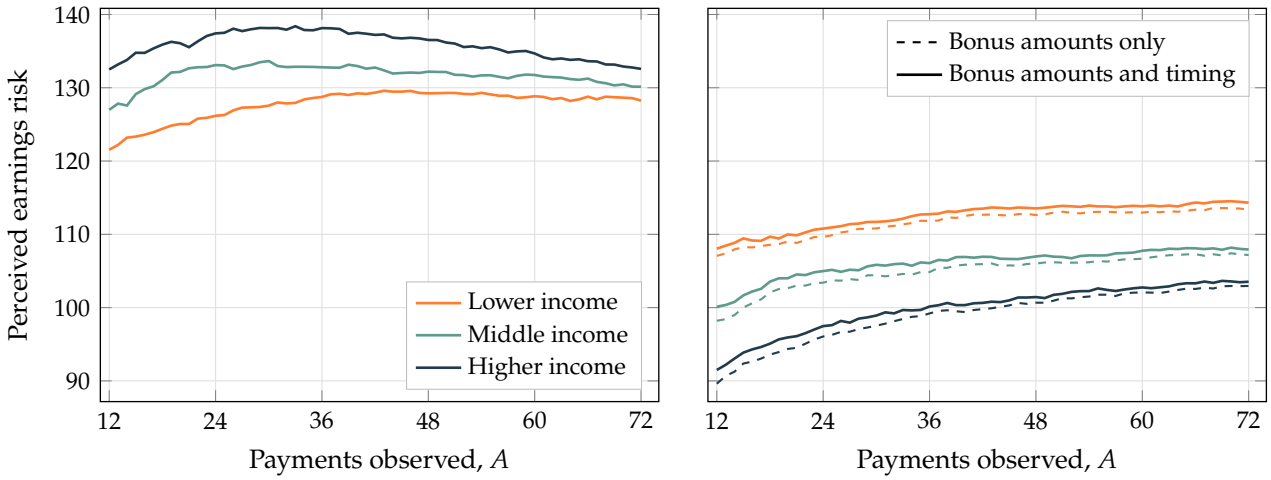
Figure J.1: Perceived risk around a scheduled annual payment

Notes: Population-weighted twelve-month forecast variance for 3,329 annual-payment recipients, in squared 100-log-point units. Month zero is the scheduled payment between payments 42 and 53. Panels share workers grouped by earnings in the preceding twelve months; no trimming or smoothing.

To assess how changes in the annual-payment contribution affect measured persistence, I replace this component of each worker's twelve $\mathcal{I}^2$ reports at $A = 48$ with its average over those reports. This preserves her mean reported risk, posterior beliefs and all other forecast components. I also retain the

workers selected by trimming the original reports, their income groups, population weights and the same resampling indices used for GMM. Persistence rises to 0.299, 0.359 and 0.362, compared with 0.282, 0.244 and 0.179 in the original series. This calculation changes the reports after learning has taken place, holding the worker's information and learning problem fixed.

Figure J.2 shows the mean-risk levels underlying the ratios in Figure 5. Each point is the group mean immediately after payment A.



(a) Total pay only

(b) Bonus pay observed

Figure J.2: Perceived earnings risk by number of payments observed

Notes: Mean risk after payment A, in squared 100-log-point units. Groups, weights and trimming follow Figure 4; points are unsmoothed. Each worker contributes one report at $A = 48$, whereas Table J.2 averages payments 48–59.